

EES 201: Introduction to Earth Sciences **(2nd year Pre-major course)**

They look at mud and see mountains, in mountains oceans, in oceans mountains to be. They go up to some rock and figure out a story, another rock, another story, and as the stories compile through time they connect—and long case histories are constructed and written from interpreted patterns of clues. This is detective work on a scale unimaginable to most detectives, with the notable exception of Sherlock Holmes.

John McPhee (American writer, Pulitzer prize winner, currently Professor of Journalism at the Princeton University)

Grading scheme:

1st Mid sem: 25 marks

2nd Mid sem: 25 marks

End sem: 50 marks

Absolute grading policy – 80 or above “A”; 79 to 65 “B”; 64 to 45 “C”; 44 to 30 “D”; < 30
“F”

Objective questions, Subjective questions, MCQs, Fill in the blanks, Pictorial questions...

References:

Grotzinger and Jordan, Understanding Earth R. Ruddiman, Earth's climate, Past Present Future, W. H. Freeman (2000), Paperback, 465 pages, ISBN 0716737418 / 9780716737414

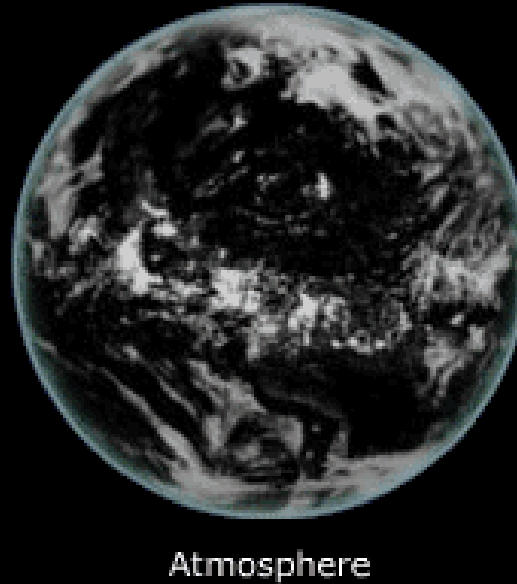
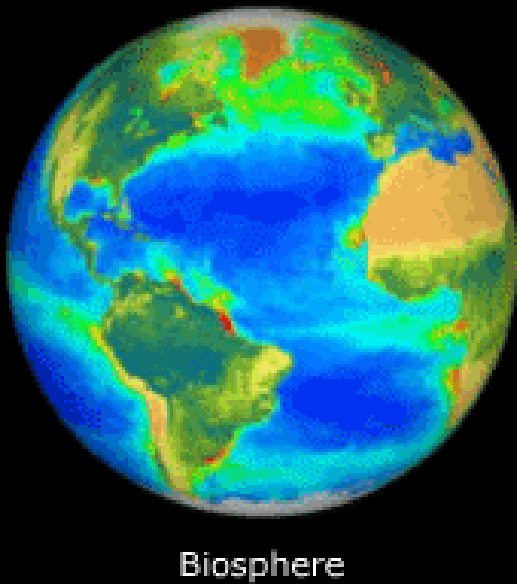
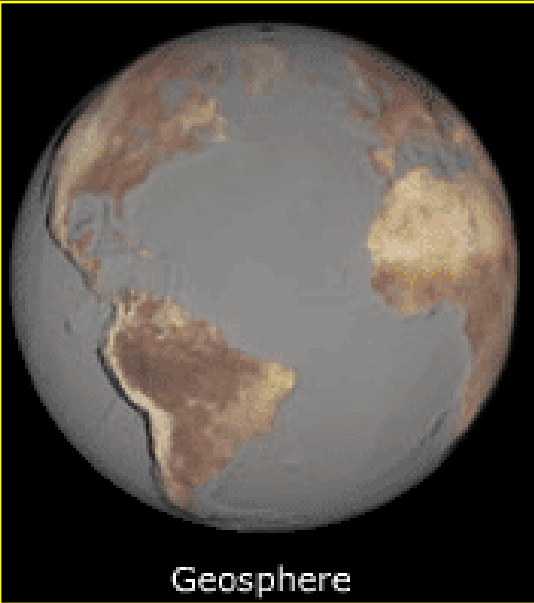
The blue marble!



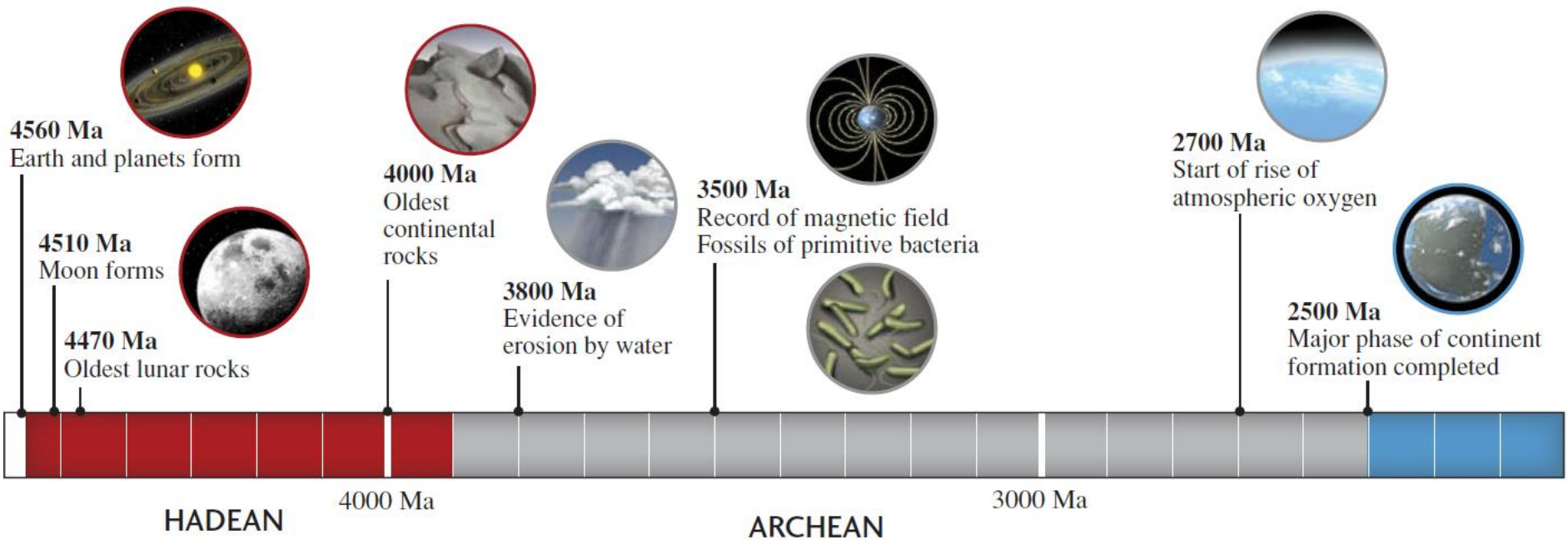
First image of the whole Earth showing the Antarctic and African continents, taken by the *Apollo 17* astronauts on December 7, 1972. [NASA.]

Astronaut Eugene A. Cernan,
commander;
Astronaut Ronald E. Evans,
command module pilot;
Scientist-astronaut Harrison H.
Schmitt, lunar module pilot

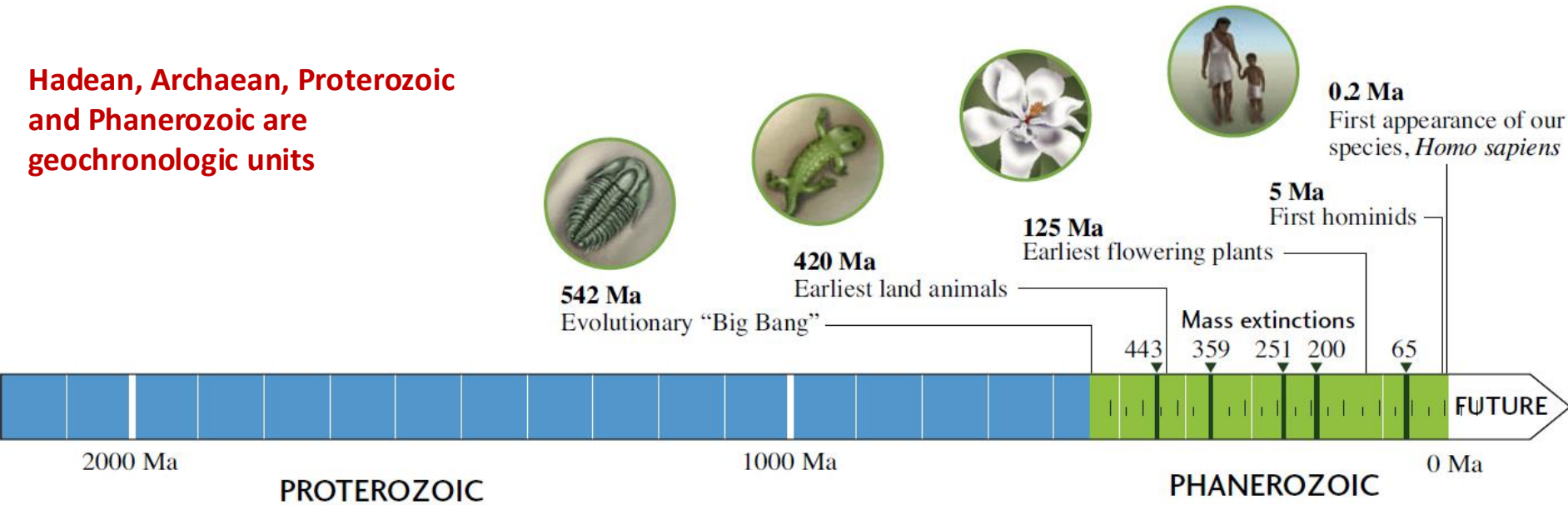
The spheres

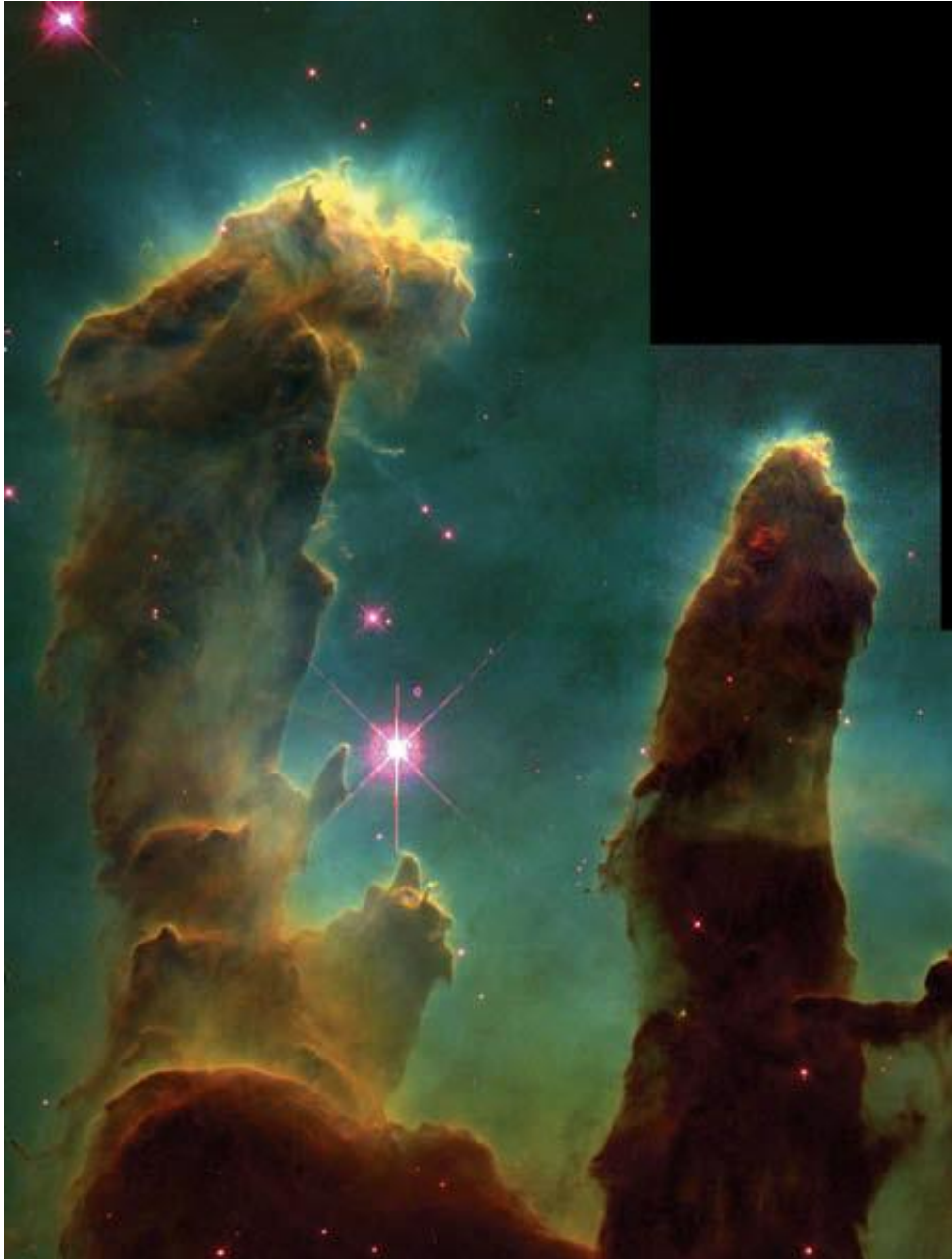


The long long history!



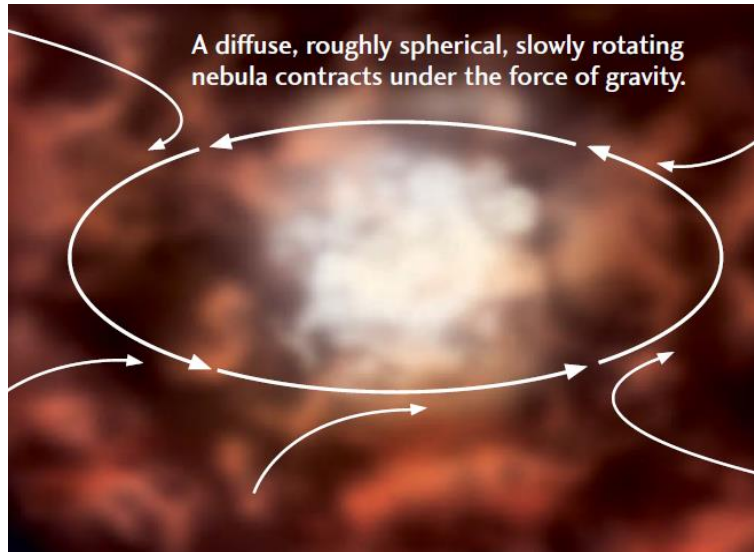
Hadean, Archaean, Proterozoic and Phanerozoic are geochronologic units



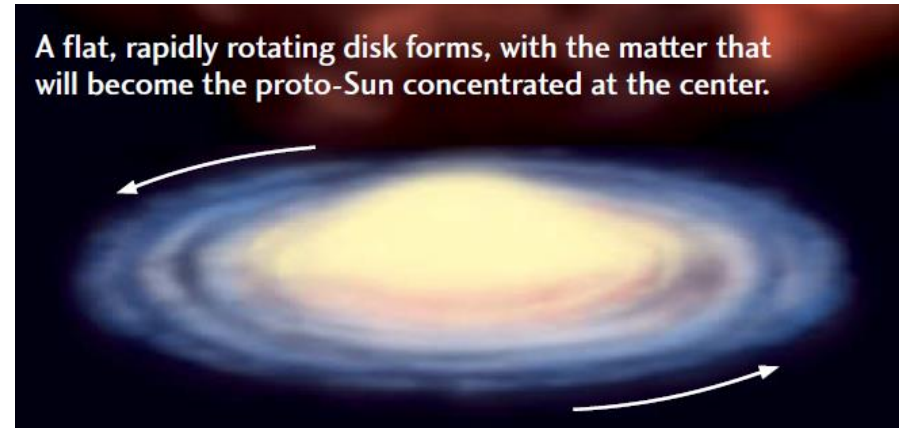


The origin?

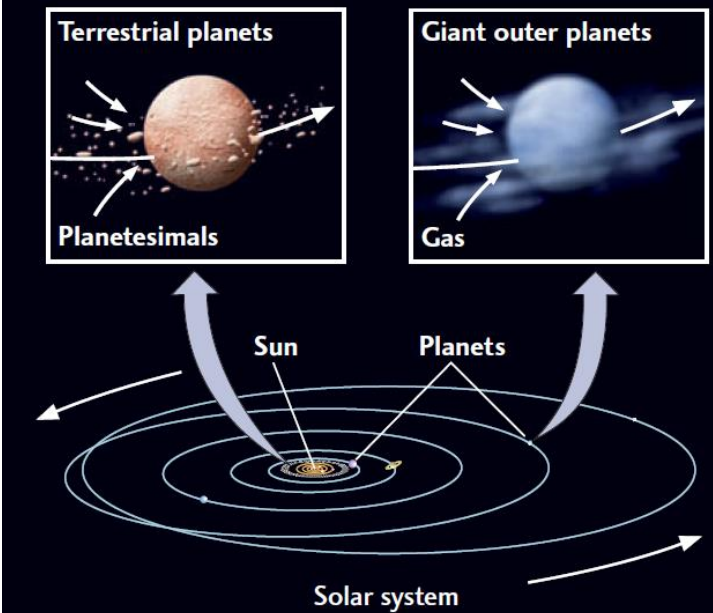
Hubble Space Telescope (in orbit around Earth) took this stunning photograph of the Eagle Nebula. The dark, pillar-like structures are columns of cool hydrogen gas and dust that give birth to new stars



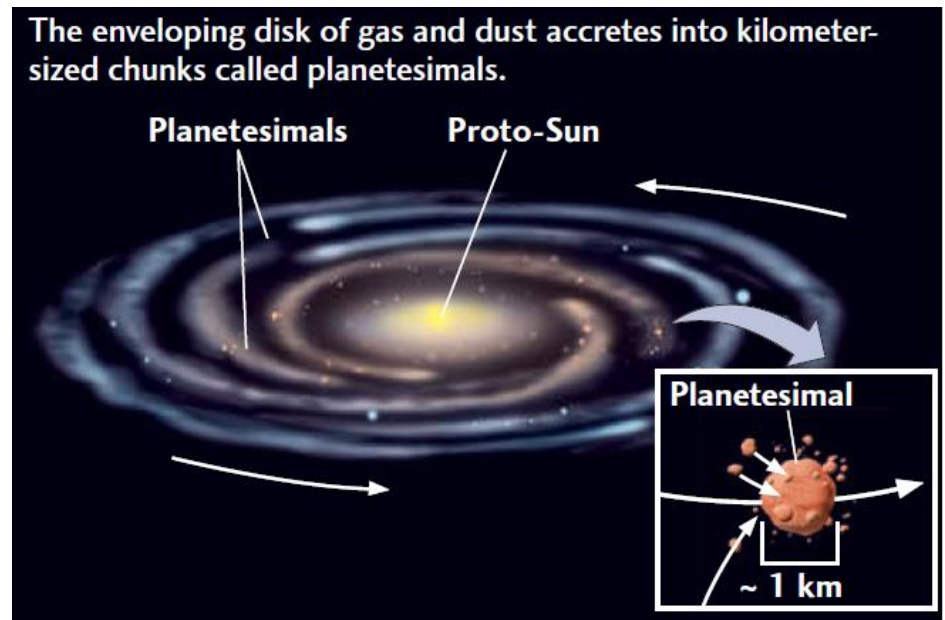
Nebular hypothesis



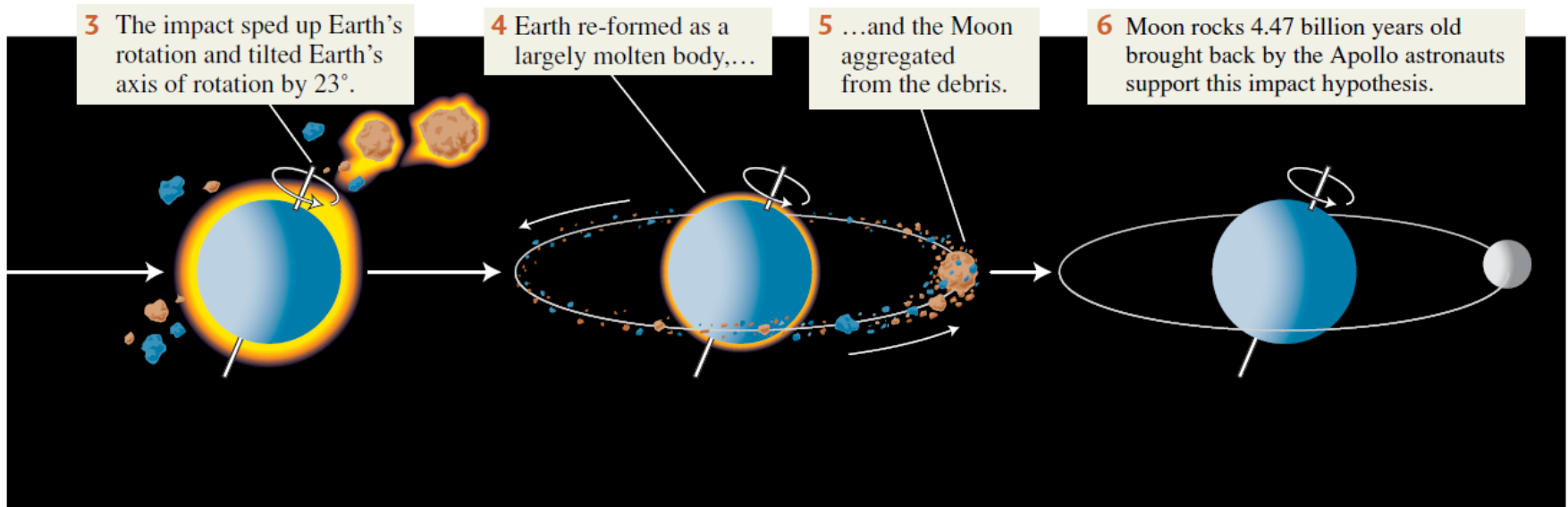
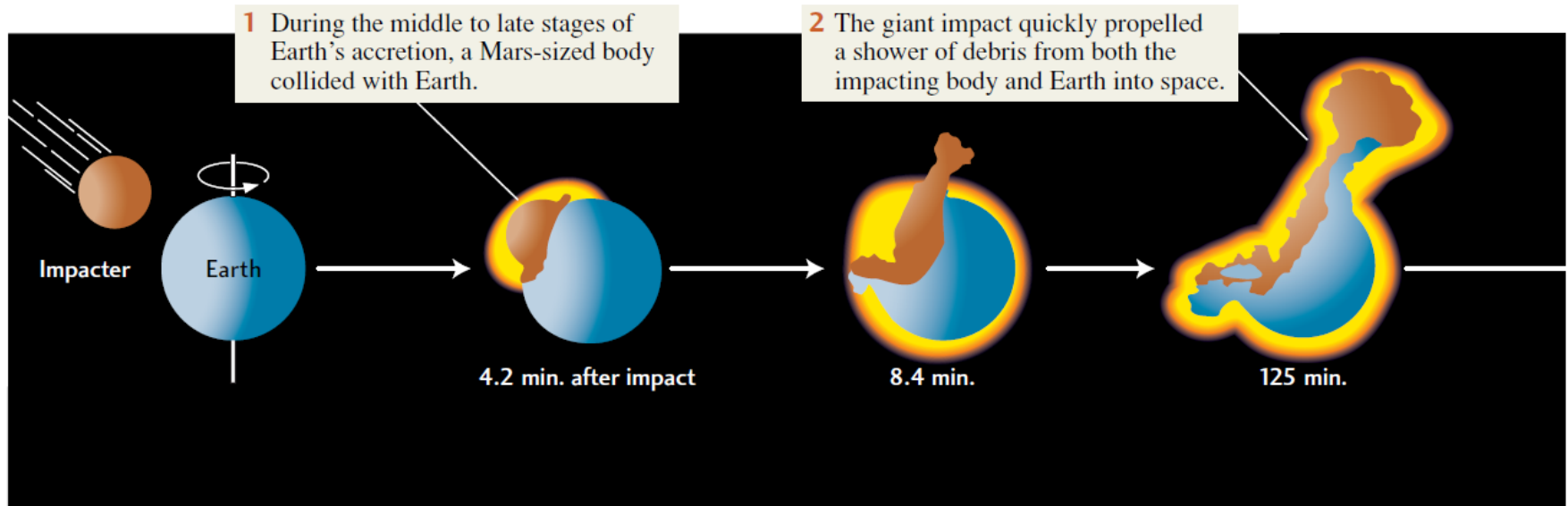
The terrestrial planets build up through collisions of planetesimals. The giant outer planets form mostly from gases.



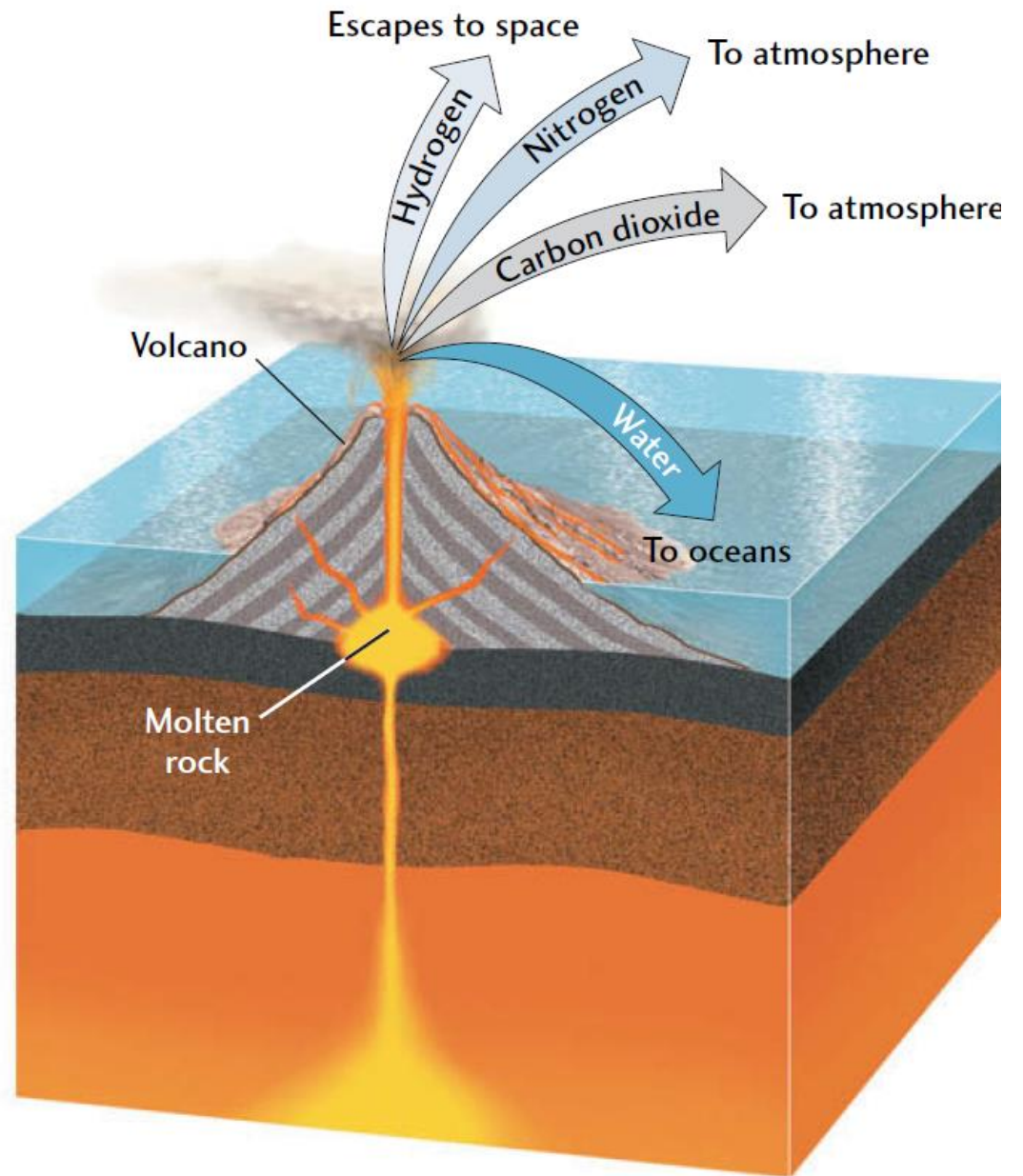
The enveloping disk of gas and dust accretes into kilometer-sized chunks called planetesimals.



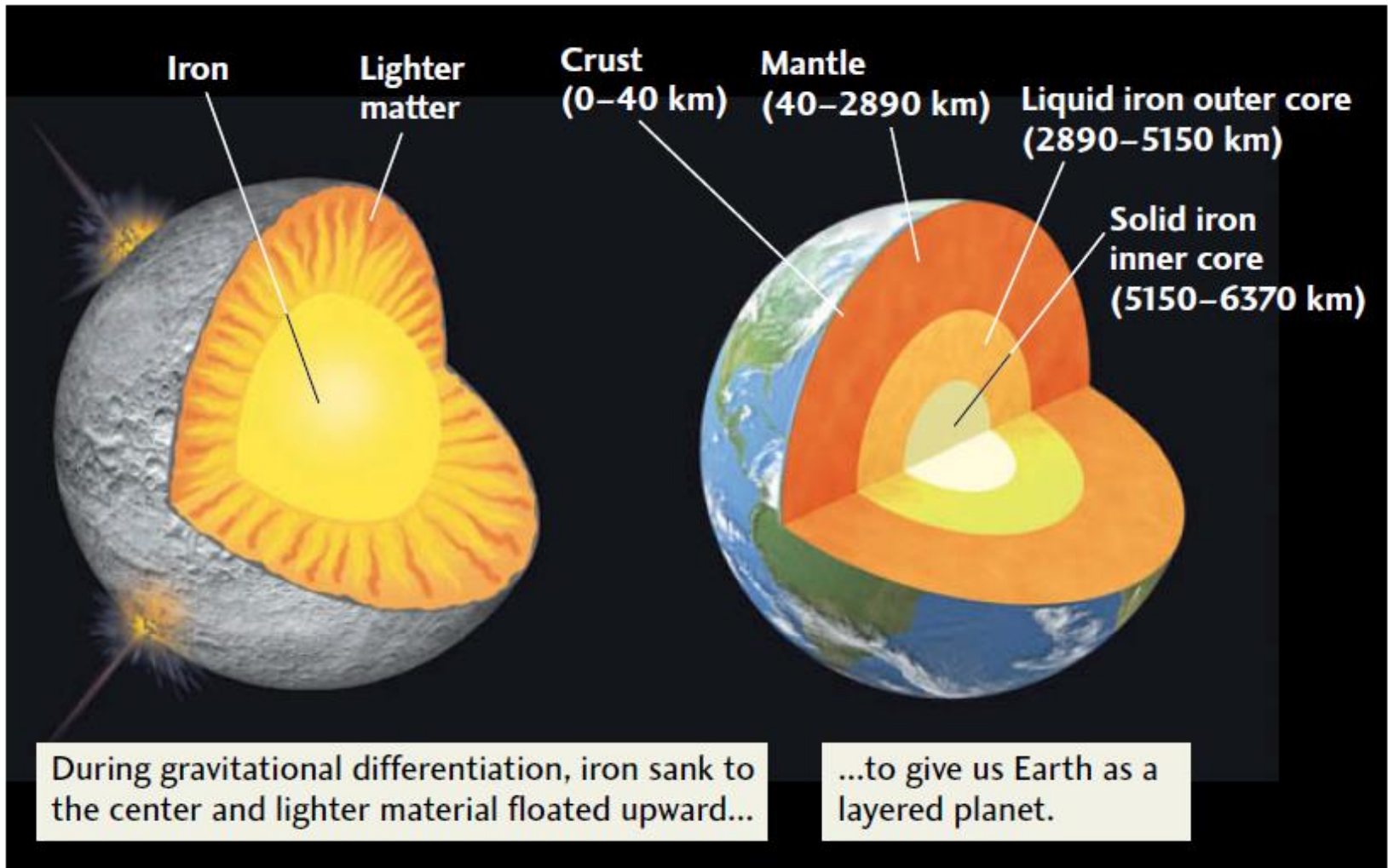
Theia?



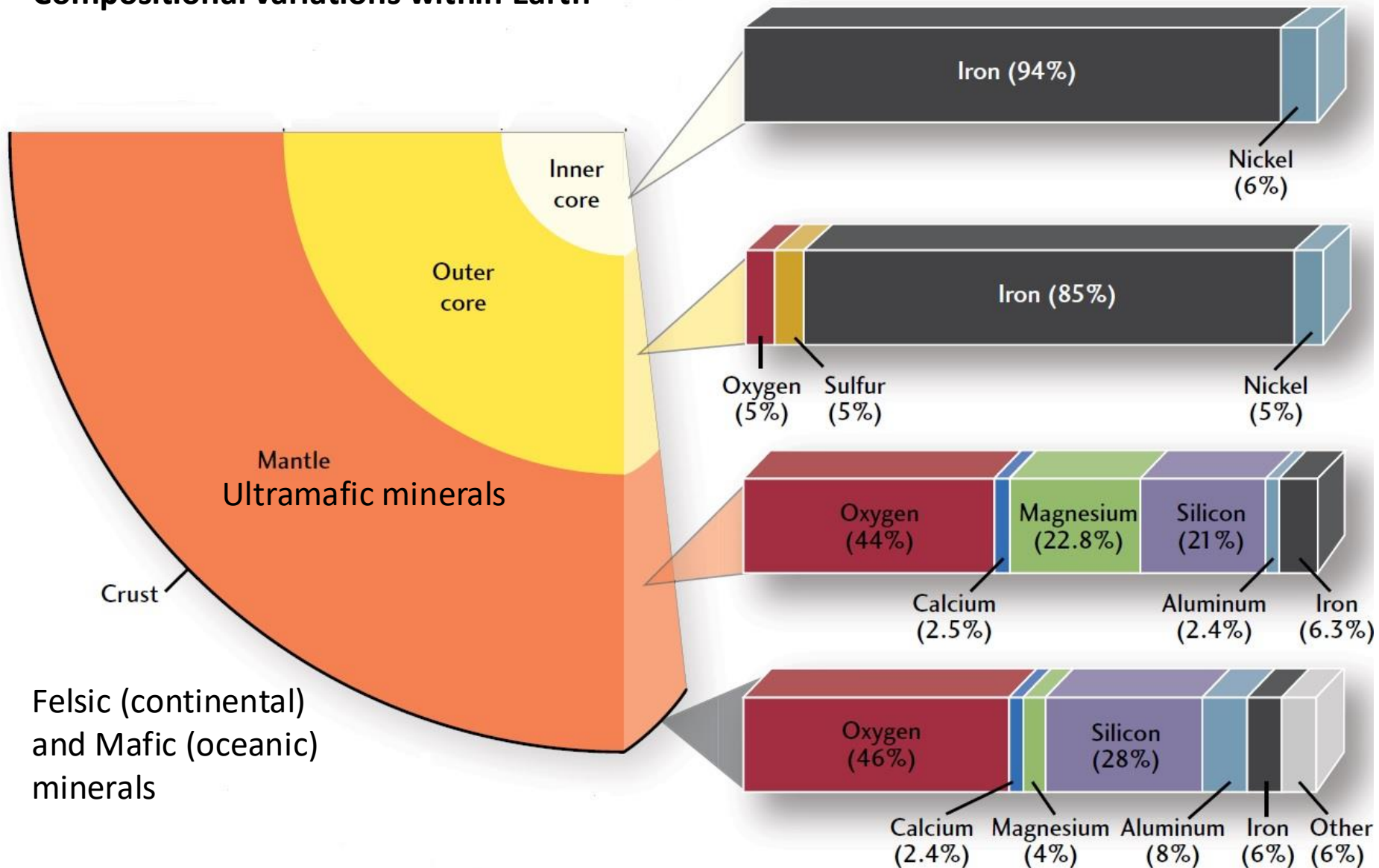
The formation of the atmosphere and hydrosphere



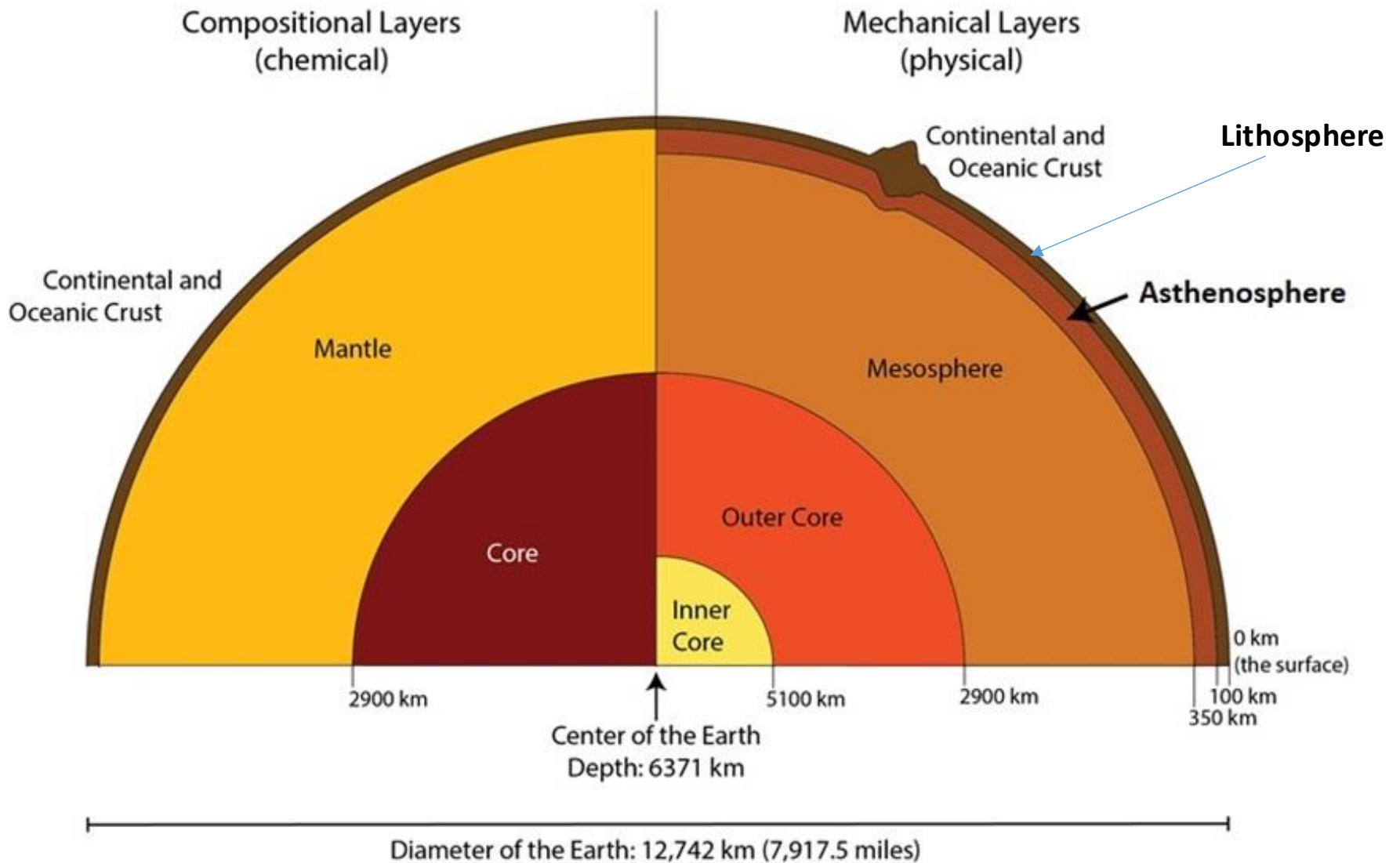
The layered structure of the Earth



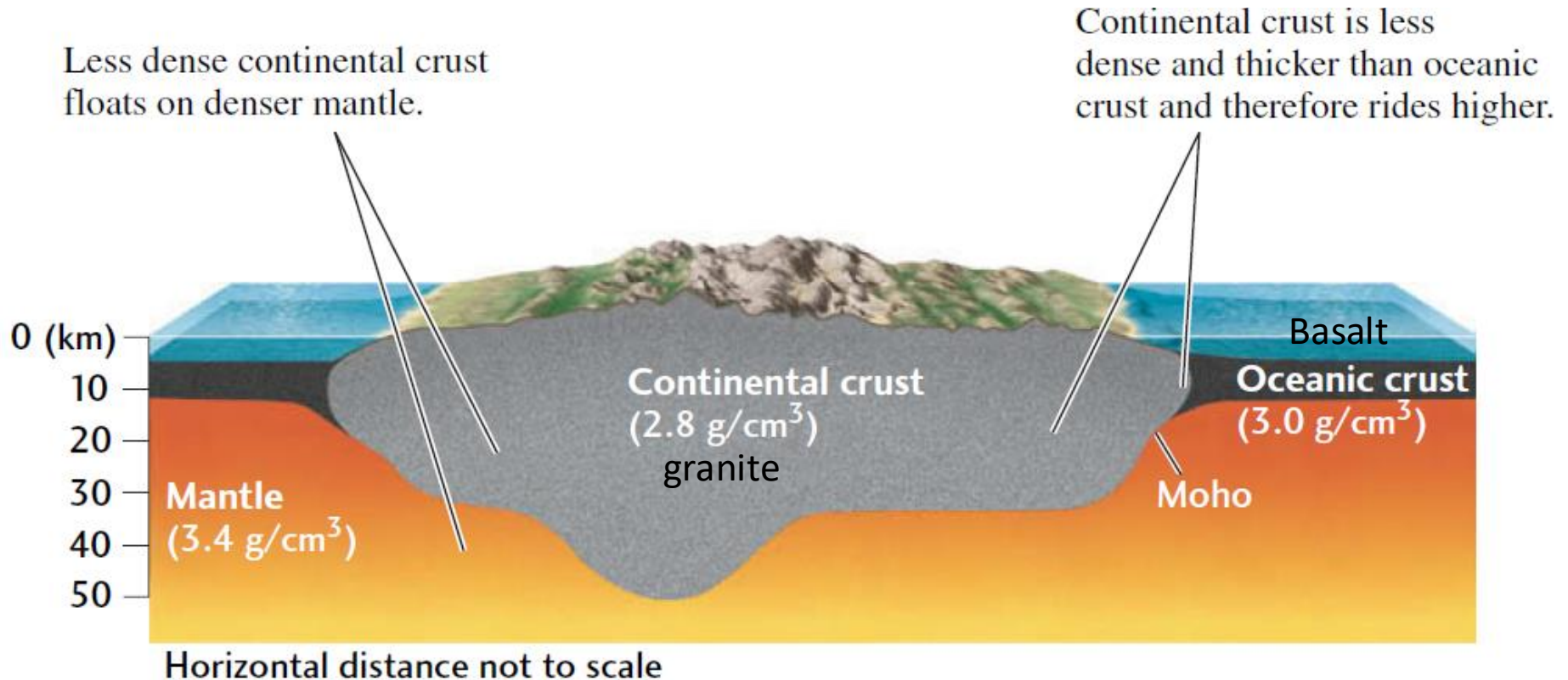
Compositional variations within Earth



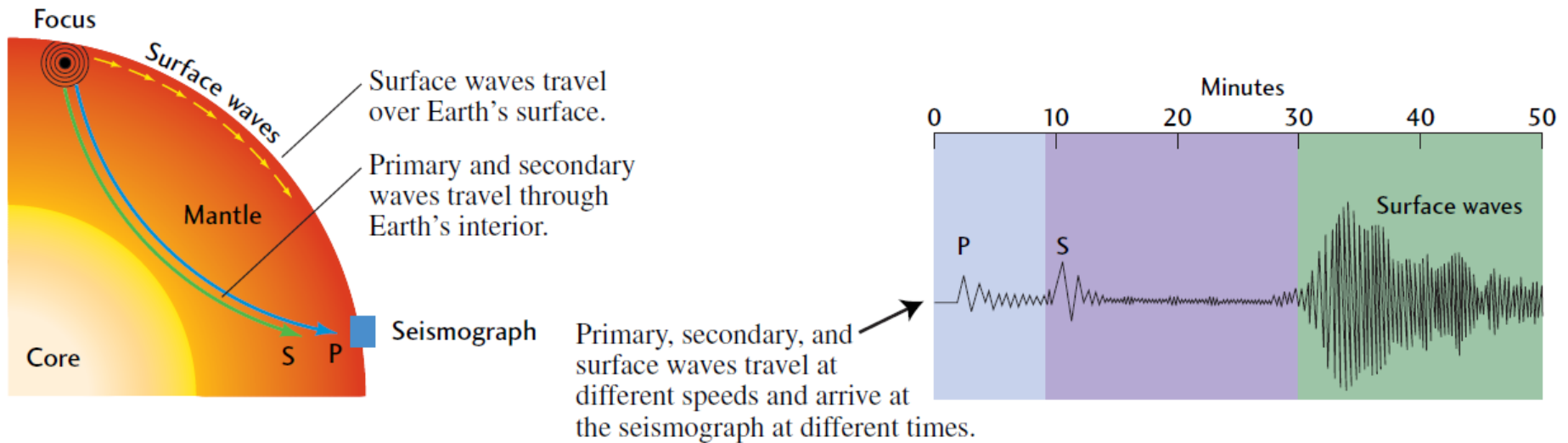
Two different views of the interior of the Earth



The outermost layer: Crust



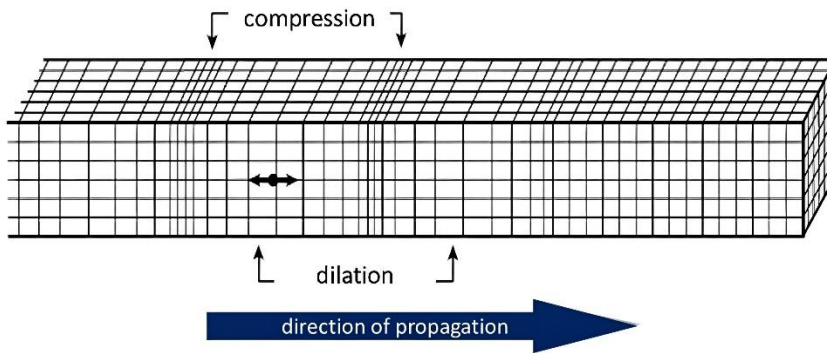
Seismic waves



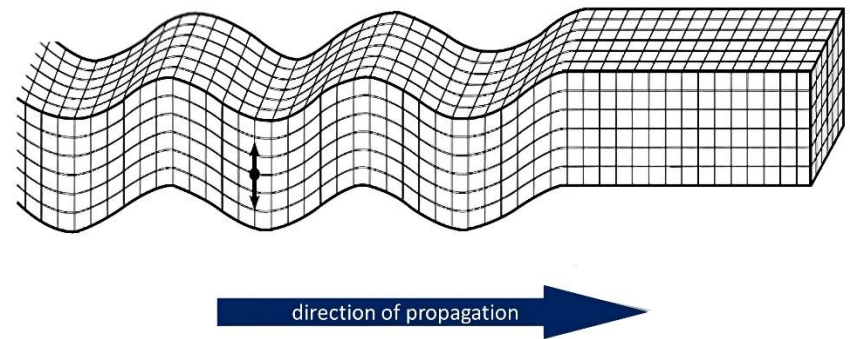
The body waves

Body Waves

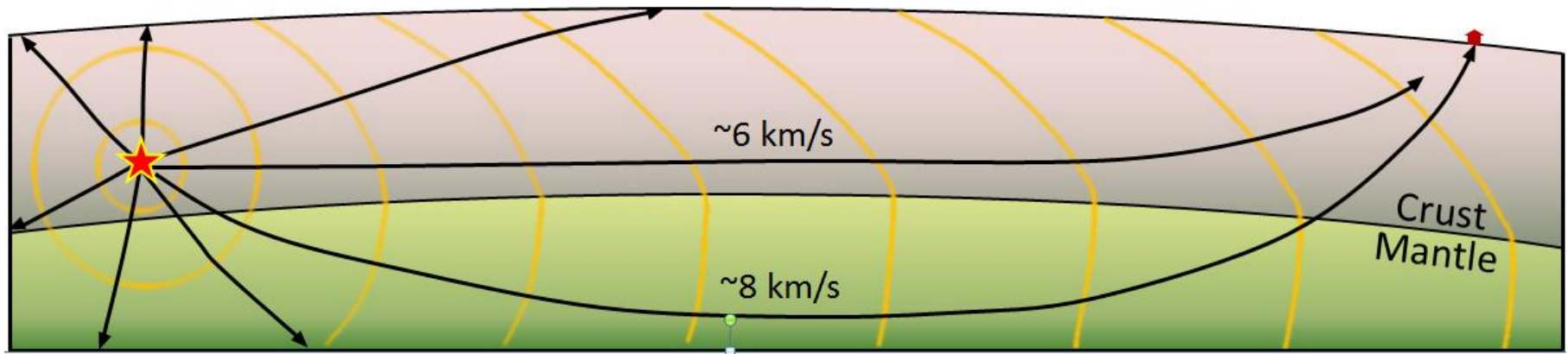
P Wave



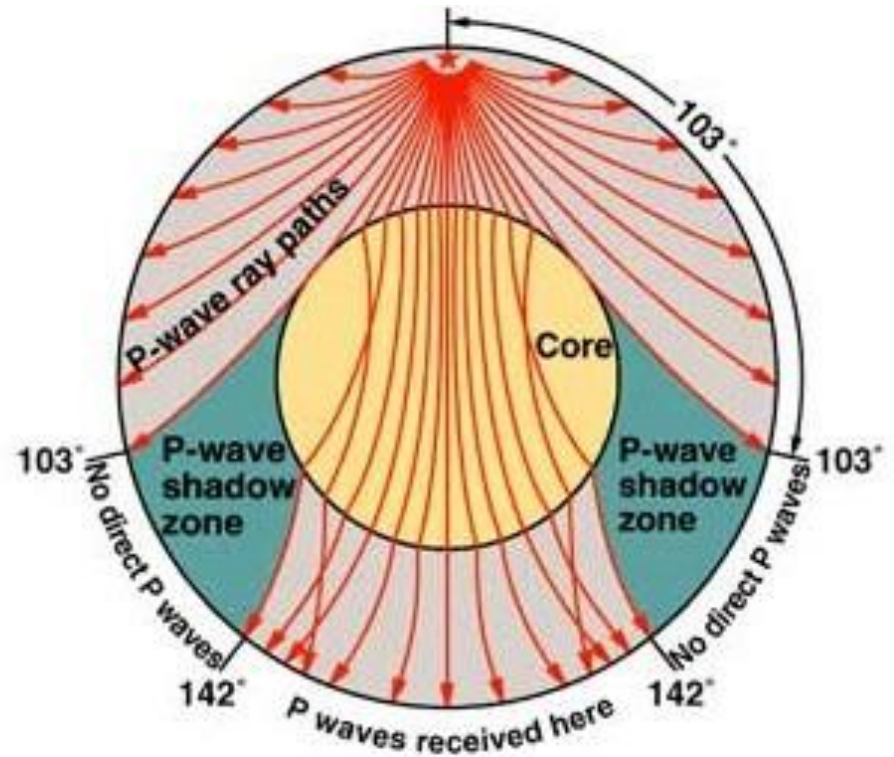
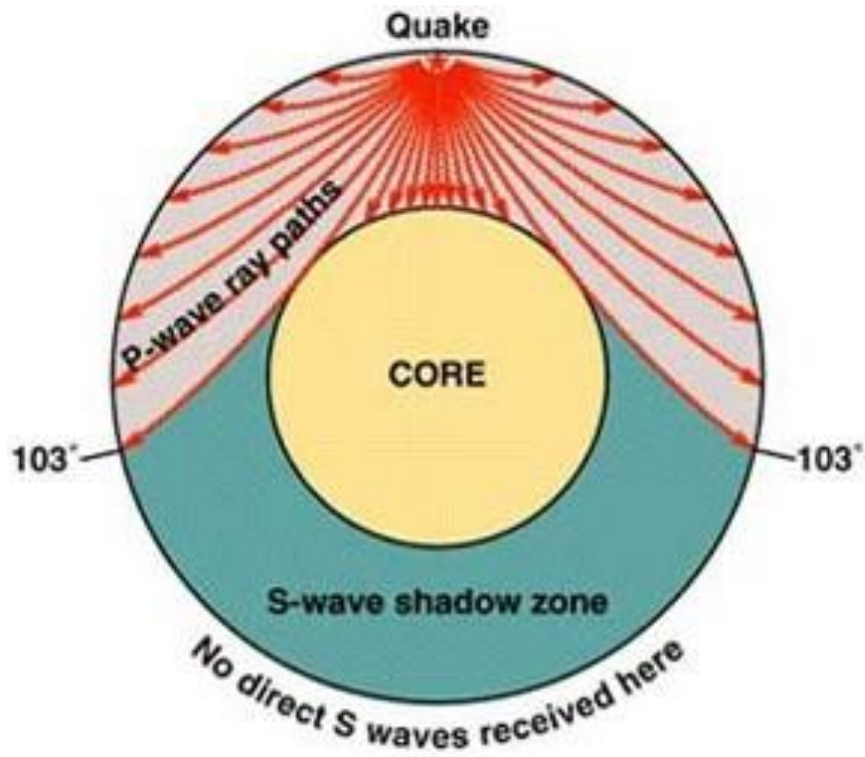
S Wave



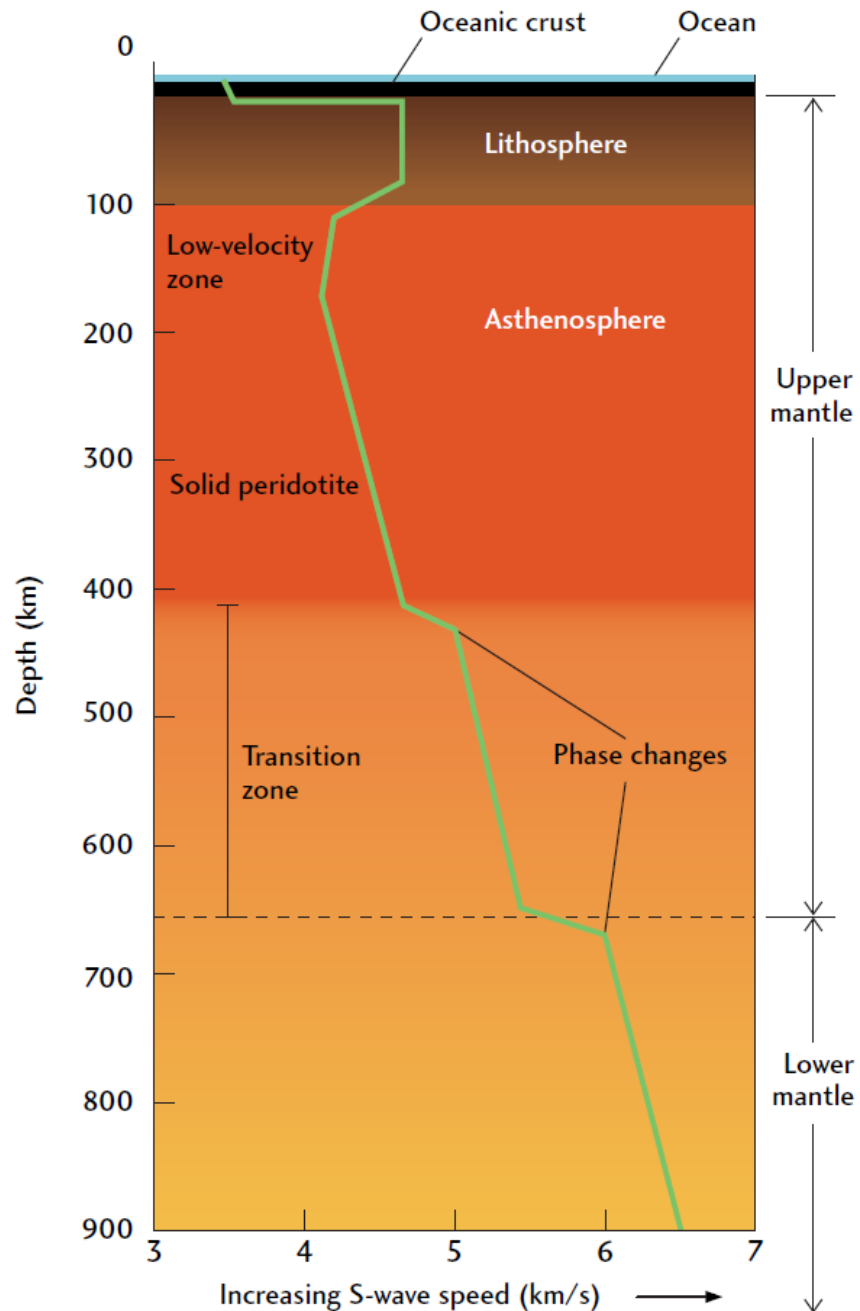
Behavior of seismic waves in the different layers of the Earth



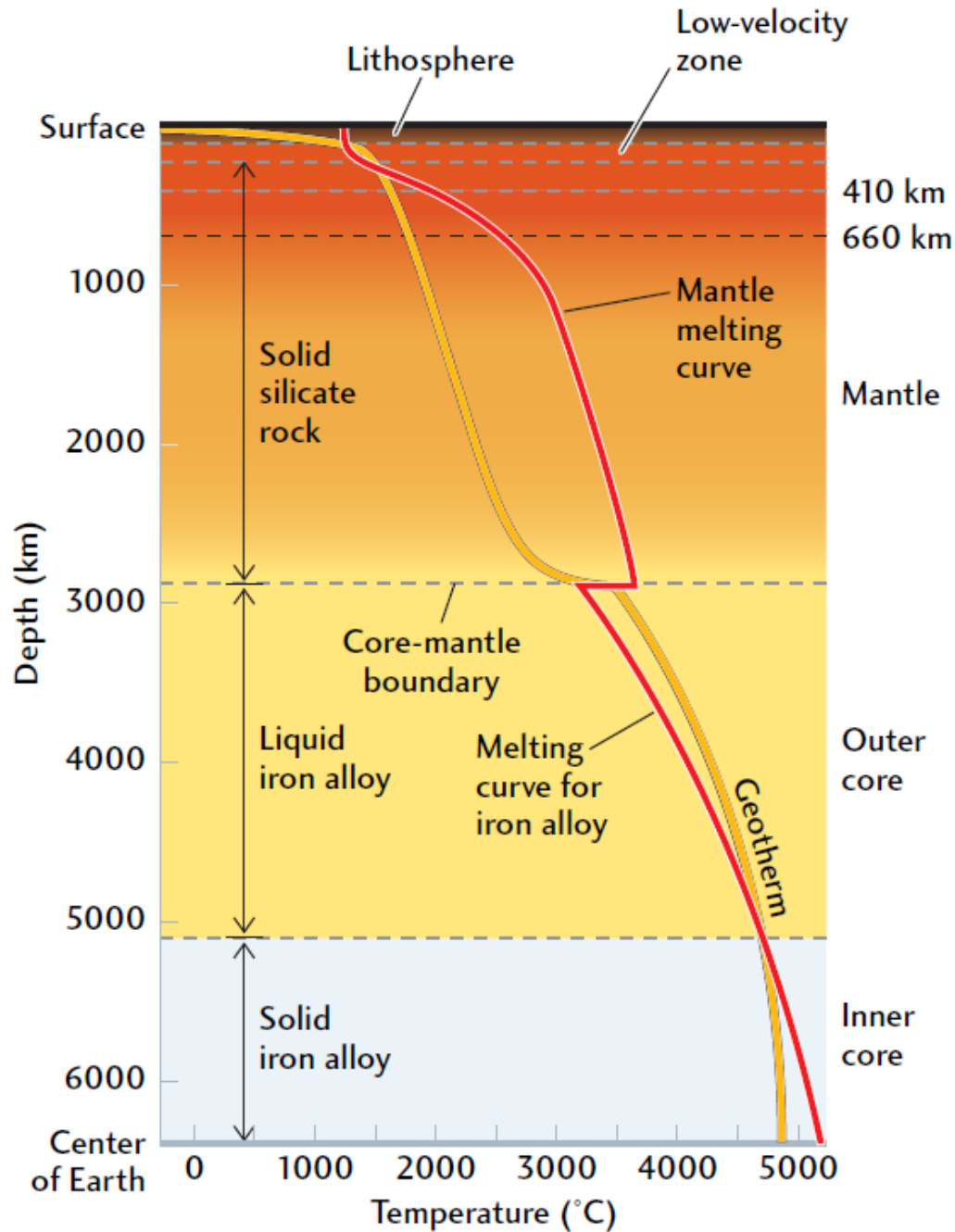
The seismic shadow zone: answer to the nature of the Earth's core



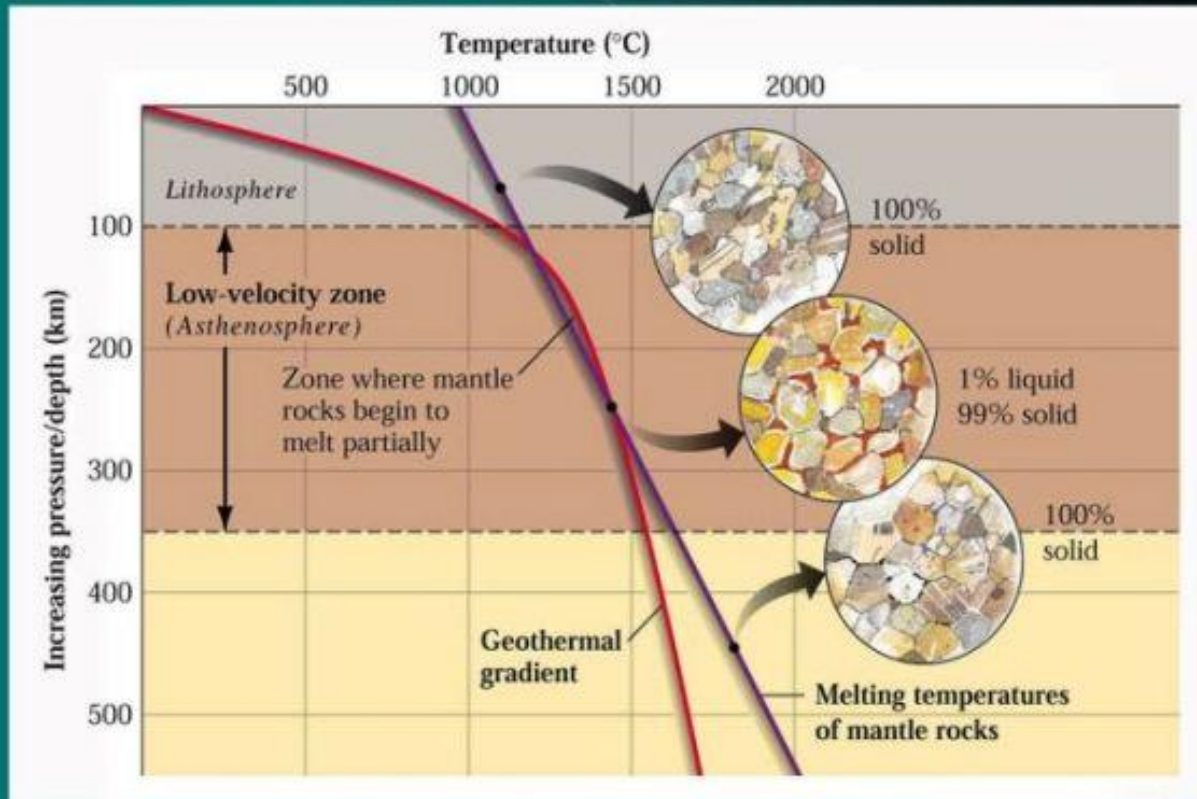
Heterogeneities in the mantle



Geothermal gradient



The low-velocity zone

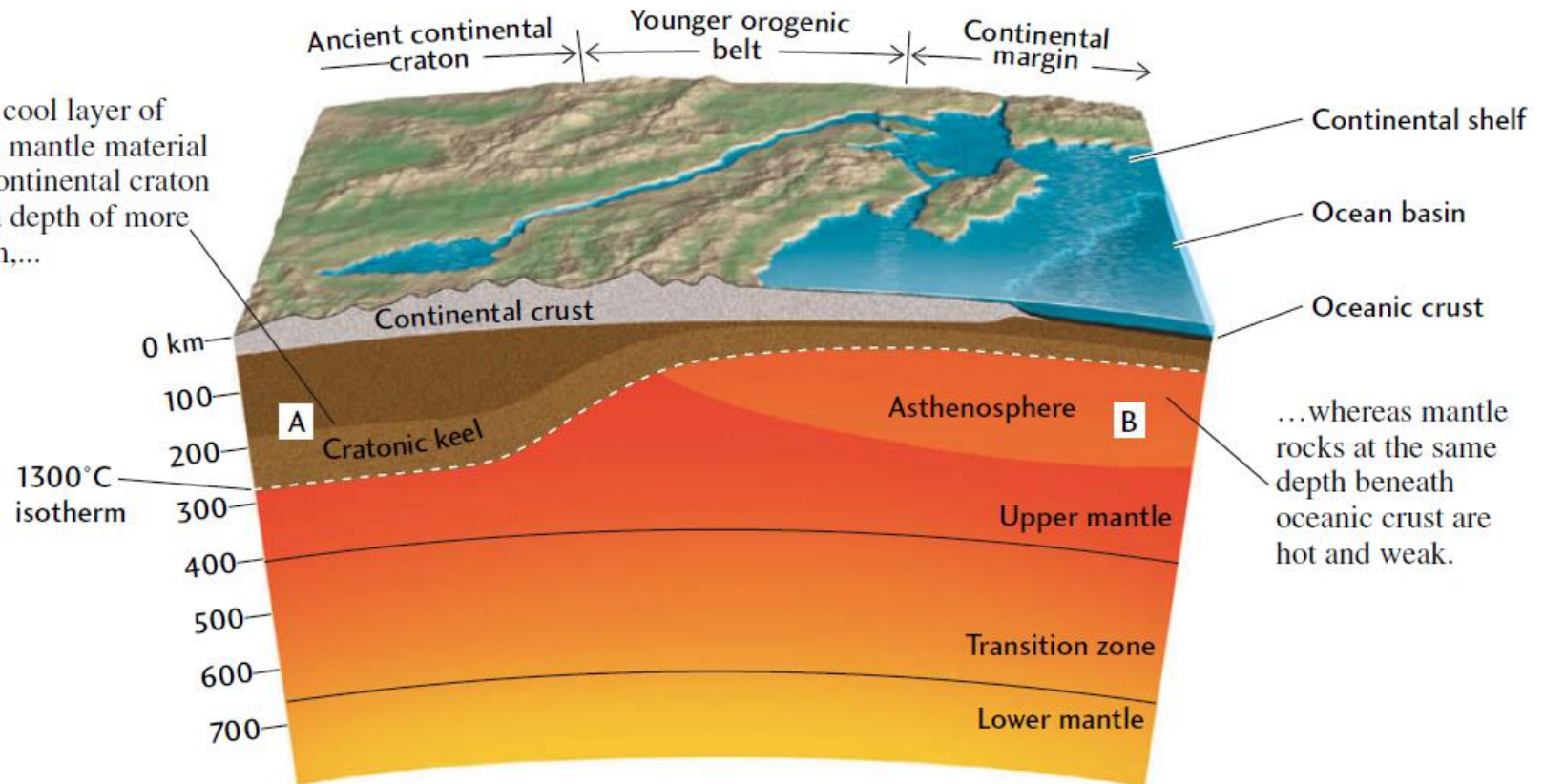


Mantle is made up of peridotite rock that is mainly composed of the mineral olivine whose main oxide composition is FeO, MgO and SiO₂.

< 1% melt distributed along a network of fissures along the grain boundaries may produce the LVZ

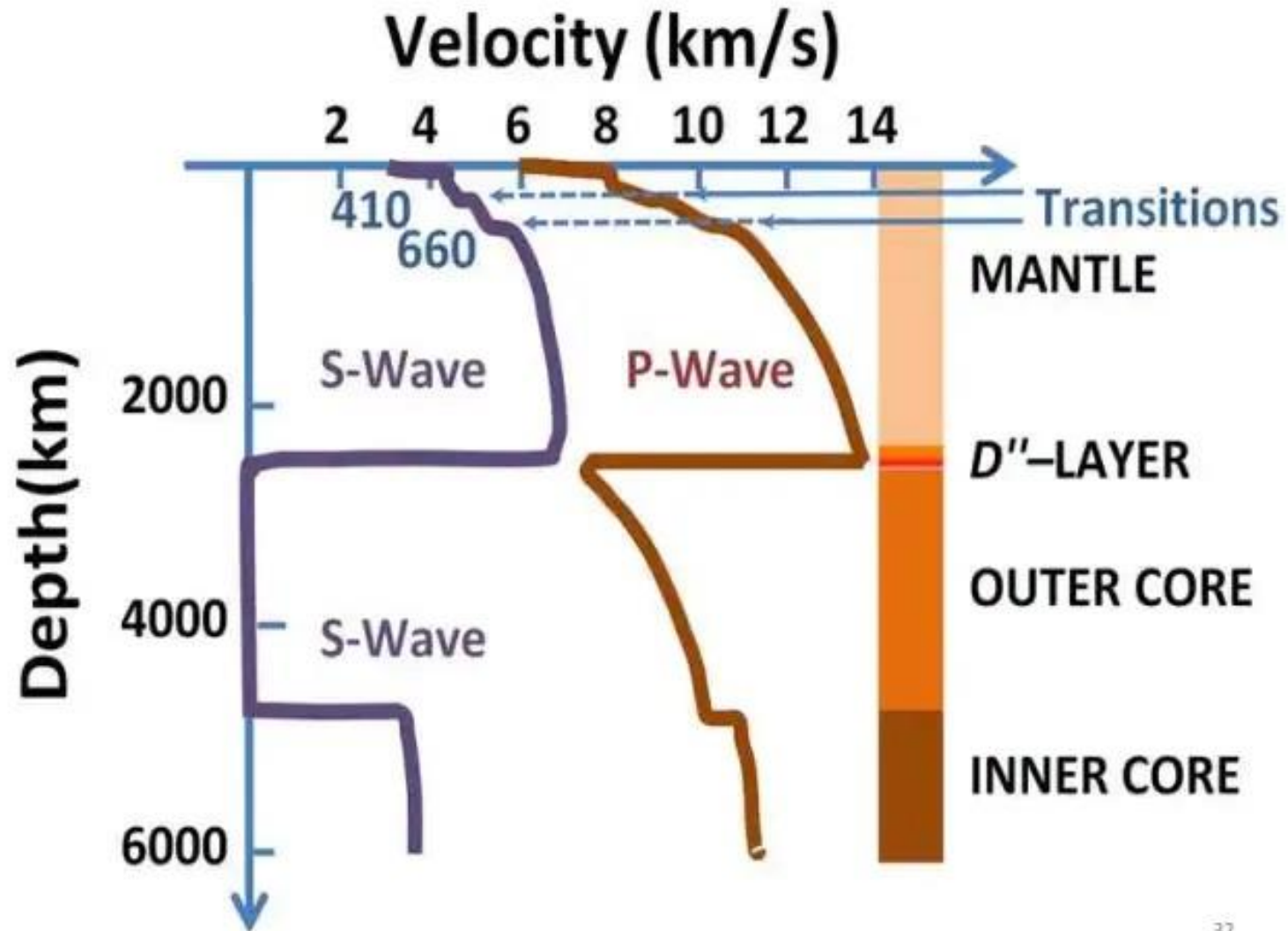
The mantle transition zone

The strong, cool layer of lithospheric mantle material beneath a continental craton extends to a depth of more than 200 km,...



Depth	Pressure	
410 km	13–14 GPa	$(\text{Mg,Fe})_2\text{SiO}_4 = (\text{Mg,Fe})_2\text{SiO}_4$ Olivine Wadsleyite (β -spinel structure)
520 km	18 GPa	$(\text{Mg,Fe})_2\text{SiO}_4 = (\text{Mg,Fe})_2\text{SiO}_4$ Wadsleyite Ringwoodite (γ -spinel structure)
660 km	23 GPa	$(\text{Mg,Fe})_2\text{SiO}_4 = (\text{Mg,Fe})\text{SiO}_3 + (\text{Mg,Fe})\text{O}$ Ringwoodite Perovskite Magnesiowüstite

The D'' layer



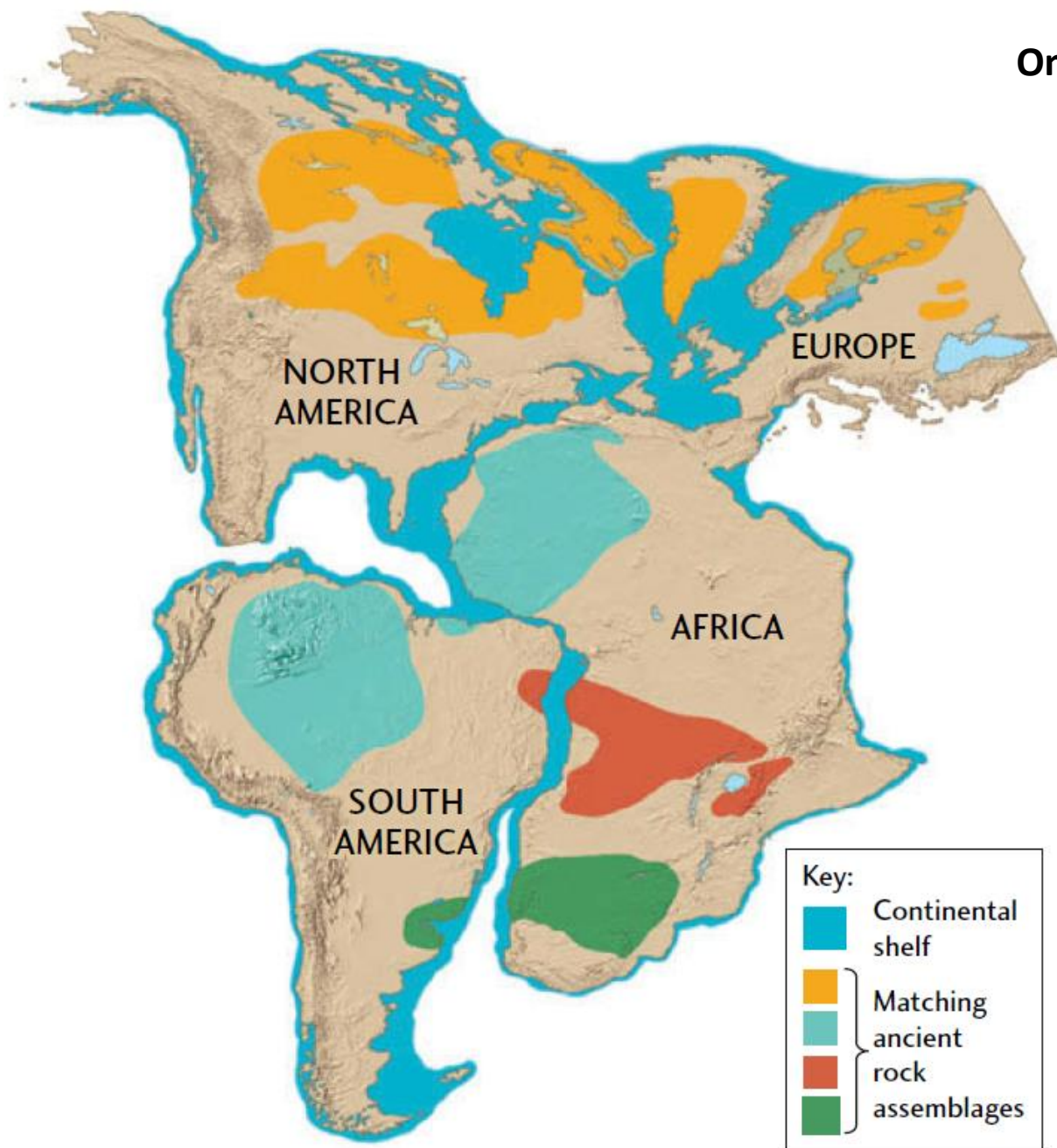
The plate tectonic system

Such changes in the superficial parts of the globe seemed to me unlikely to happen if the earth were solid to the center. I therefore imagined that the internal parts might be a fluid more dense, and of greater specific gravity than any of the solids we are acquainted with, which therefore might swim in or upon that fluid. Thus the surface of the earth would be a shell, capable of being broken and disordered by the violent movements of the fluid on which it rested.

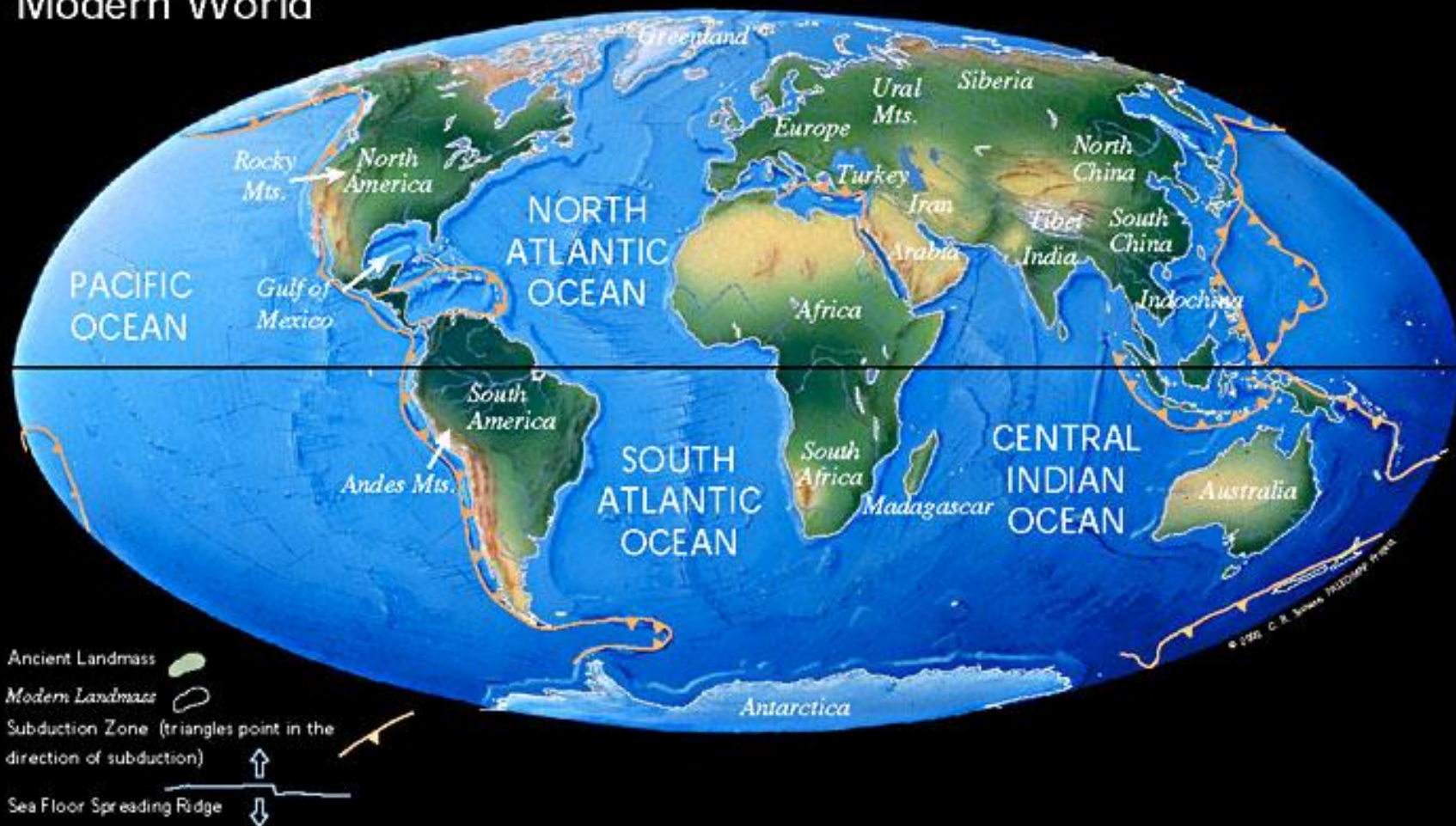
(Benjamin Franklin, 1782, in a letter to French geologist Abbé J. L. Giraud-Soulavie)



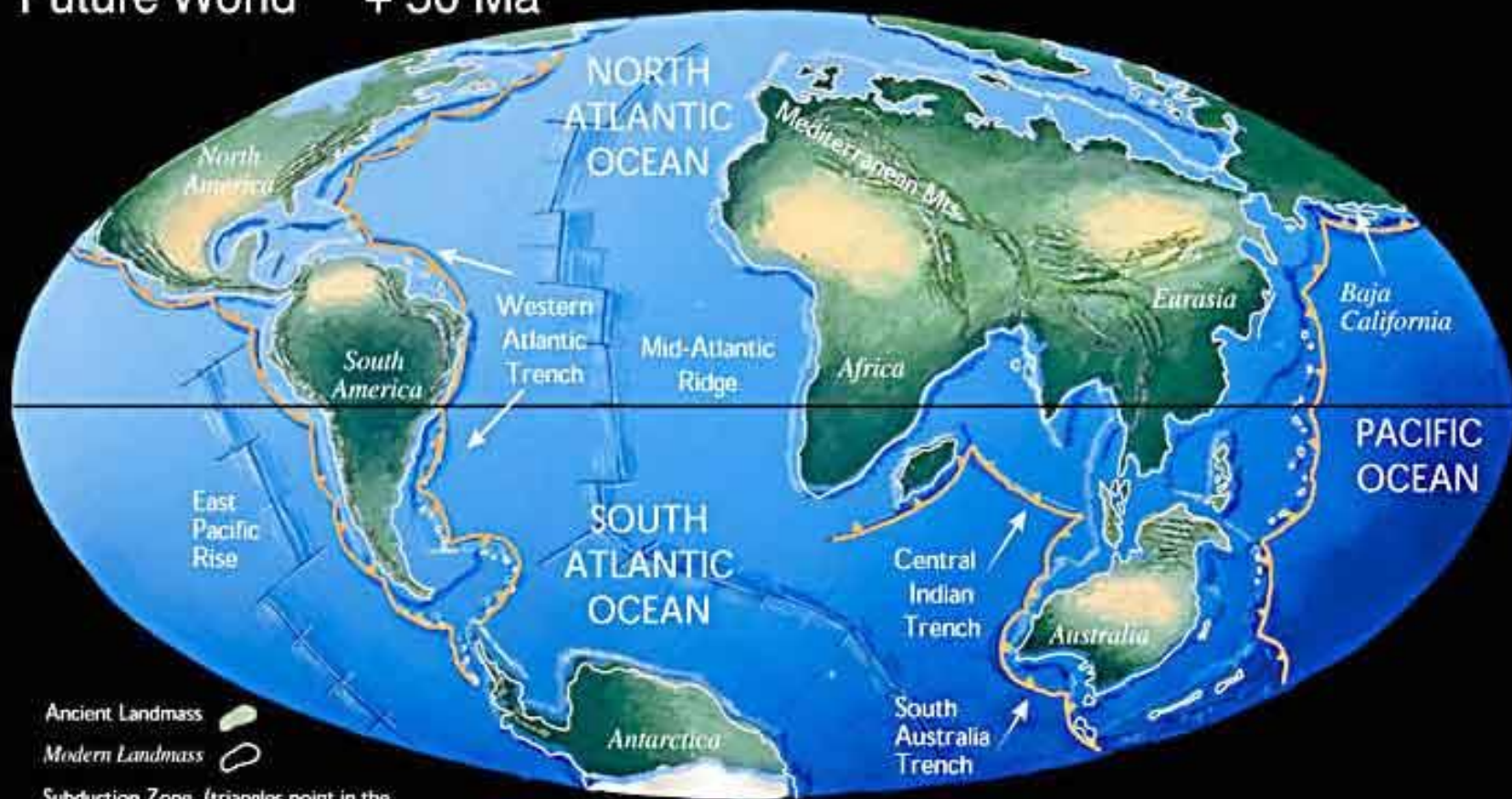
One continent in the past



Modern World



Future World + 50 Ma



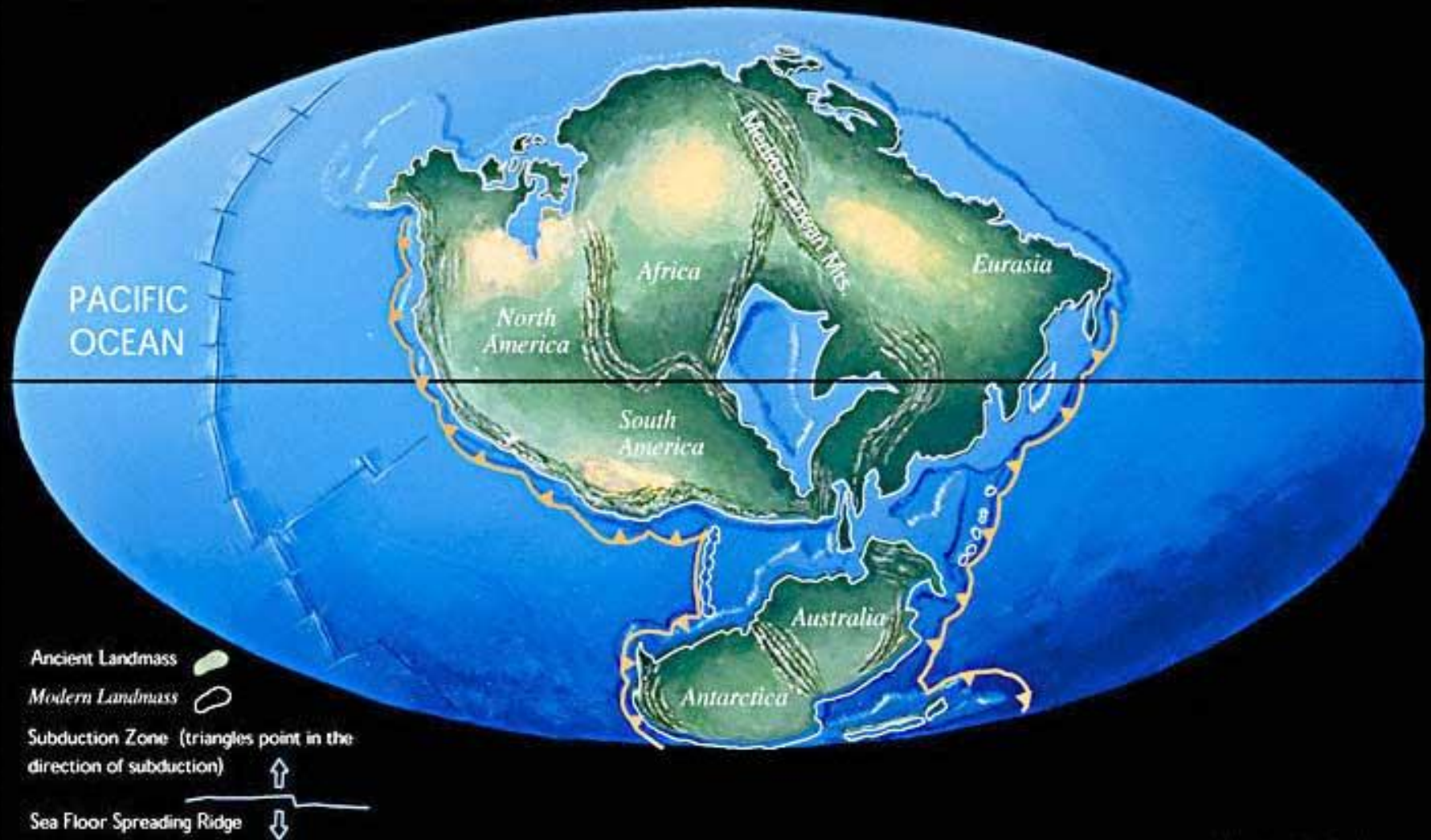
Ancient Landmass

Modern Landmass

Subduction Zone (triangles point in the direction of subduction)

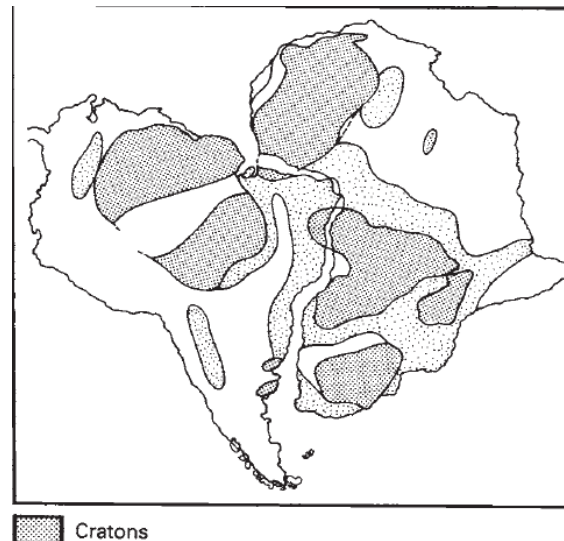
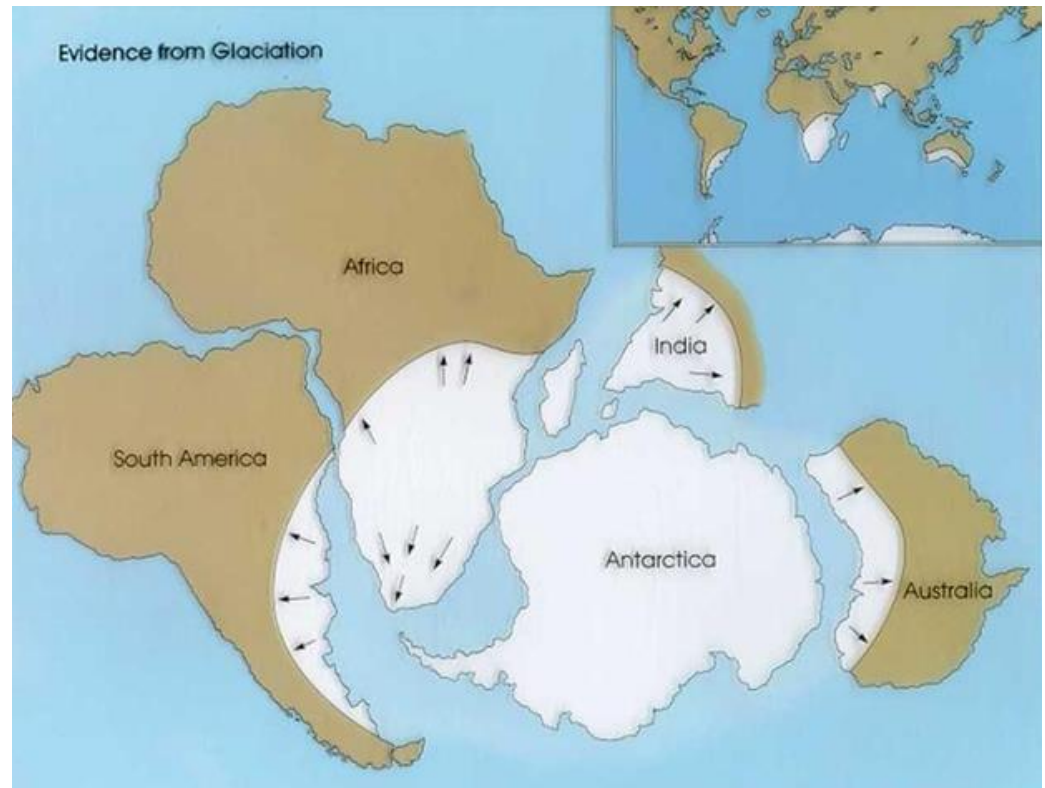
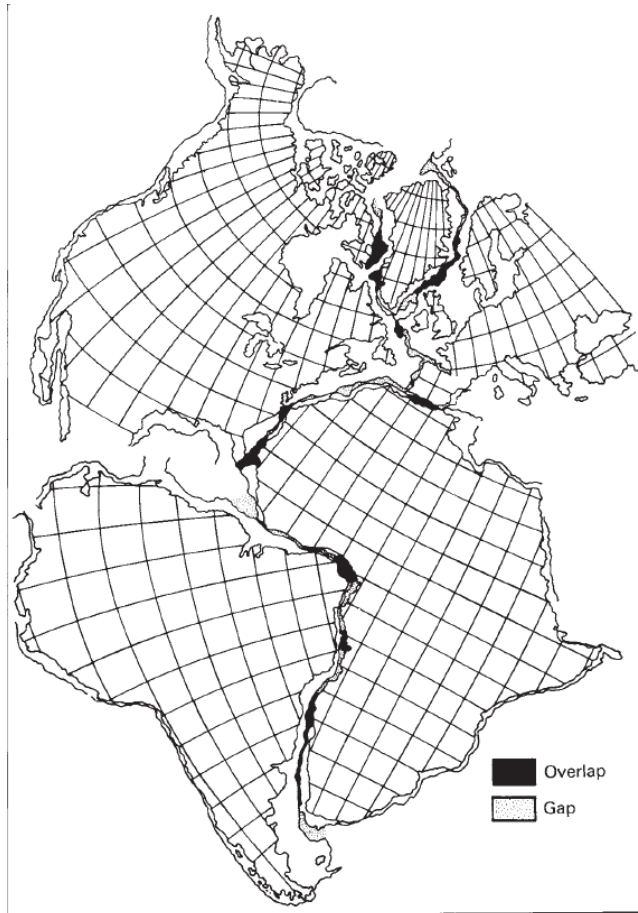
Sea Floor Spreading Ridge

Future World + 250 Ma



Evidences of plate tectonics

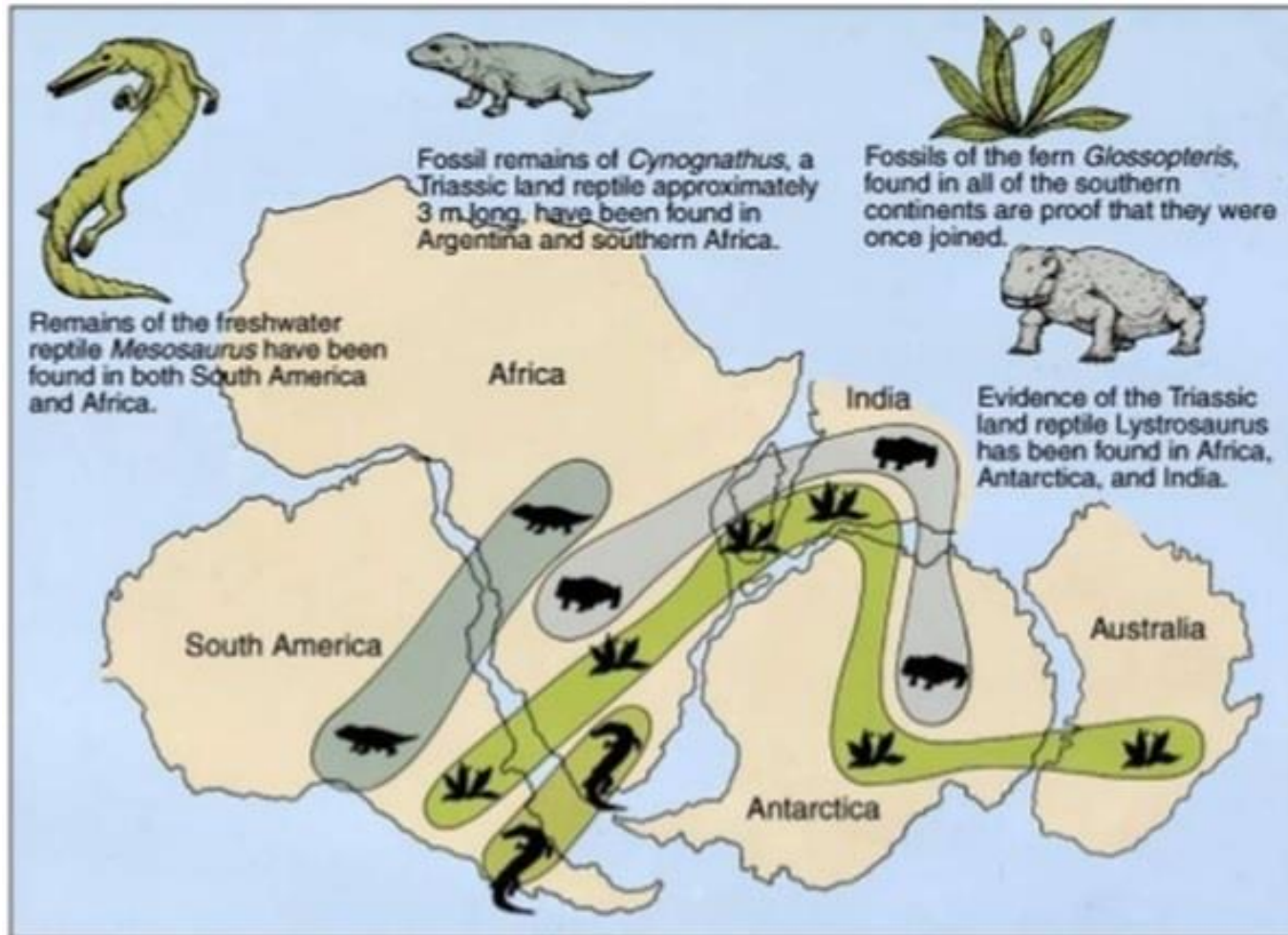
Correlatable cratons, igneous provinces, metallogenic provinces



Correlatable glaciogenic sediments

Correlatable cratons
(stable part of lithosphere)

Evidences of plate tectonics



The animals and plants on the different continents showed similarities in their evolution until the postulated breakup time. After that, they followed different evolutionary paths because of their isolation and changing environments on the separating continents.

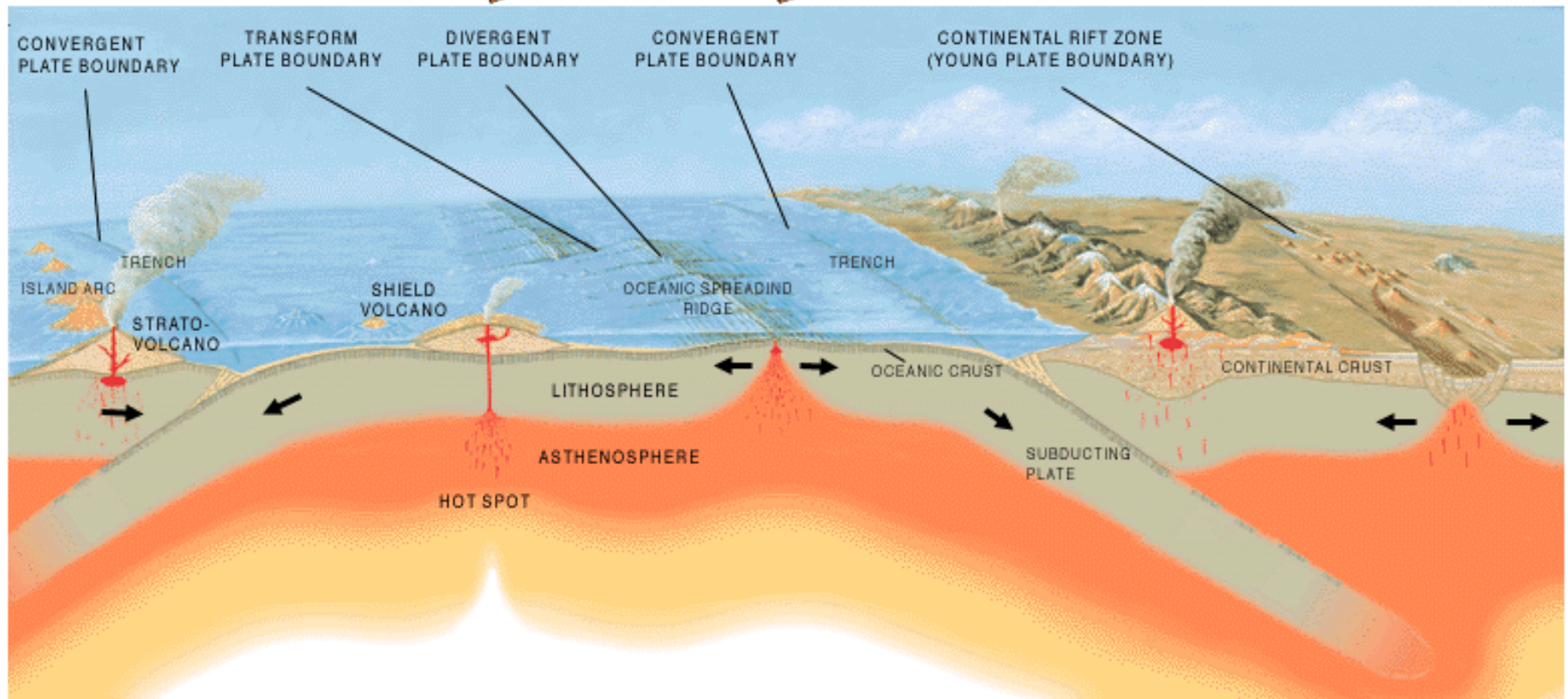
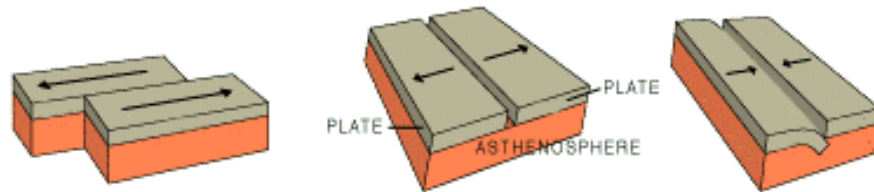
Fossils of *Lystrosaurus* from Madhukunda, West Bengal



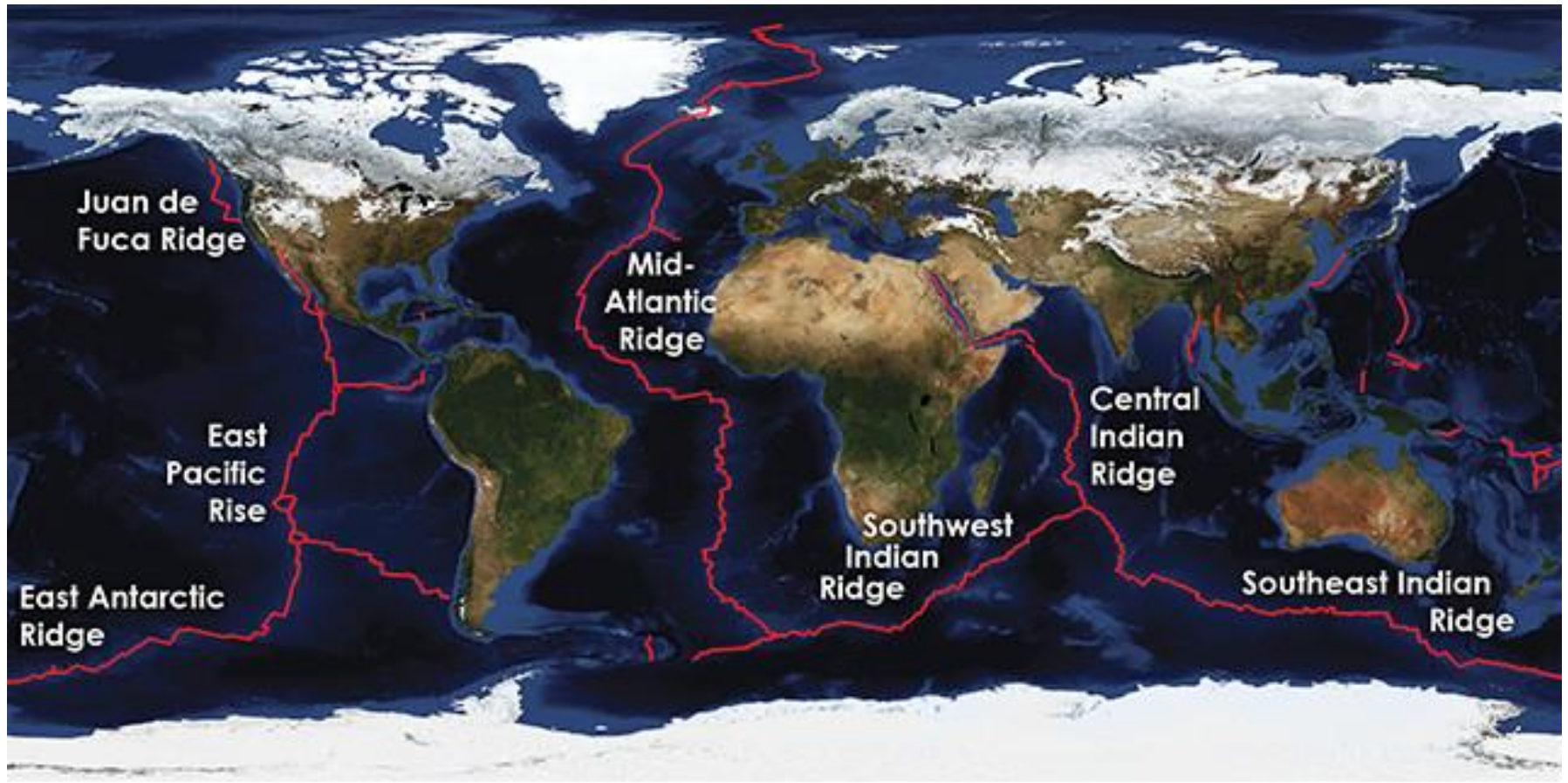


Continental drift? No!
Seafloor spreading? Yes!

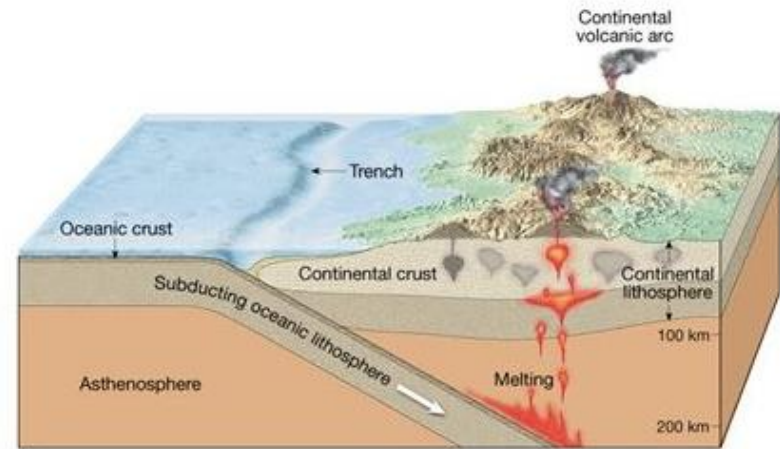
Divergent and convergent boundaries



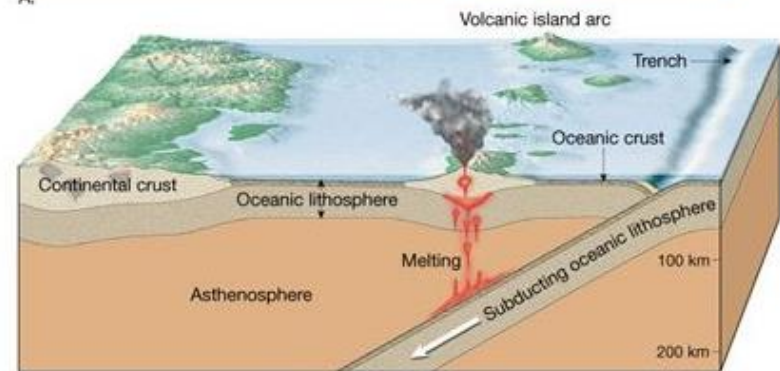
Oceanic ridges



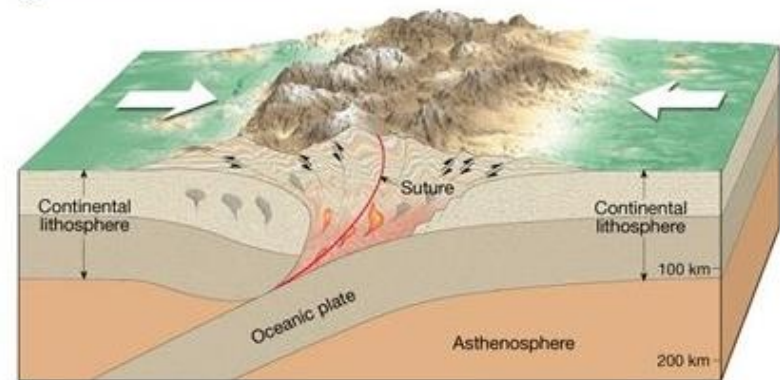
Three types of convergent boundaries



A.

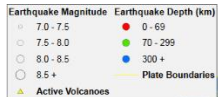
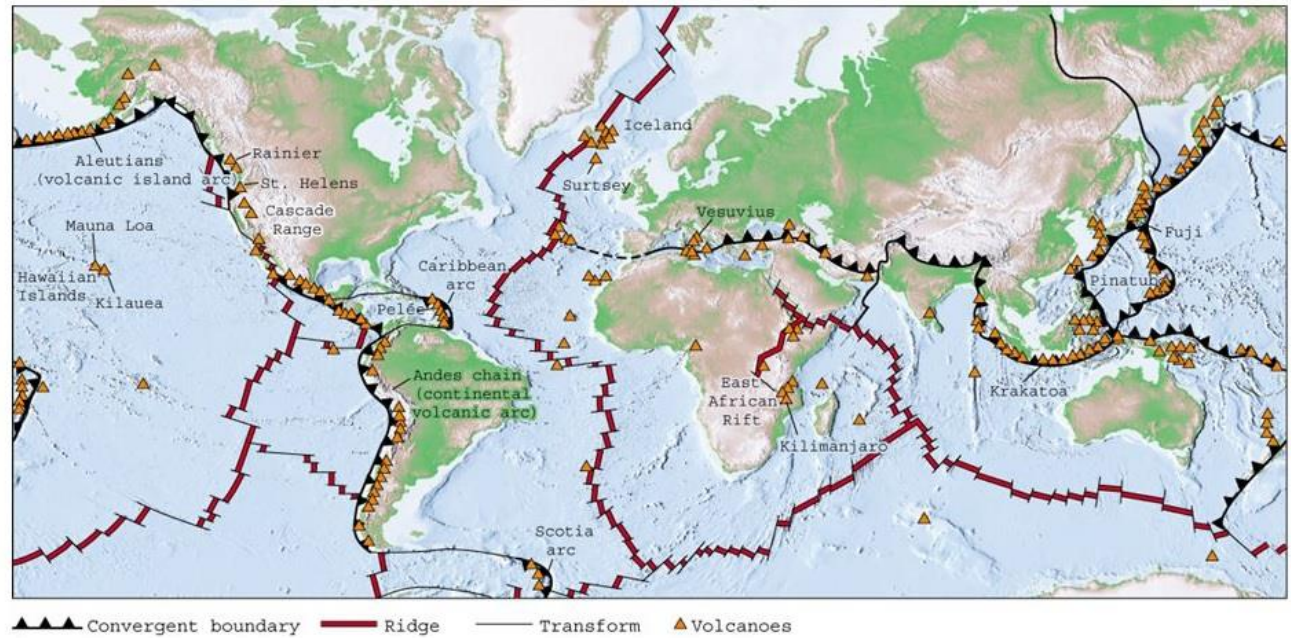


B.

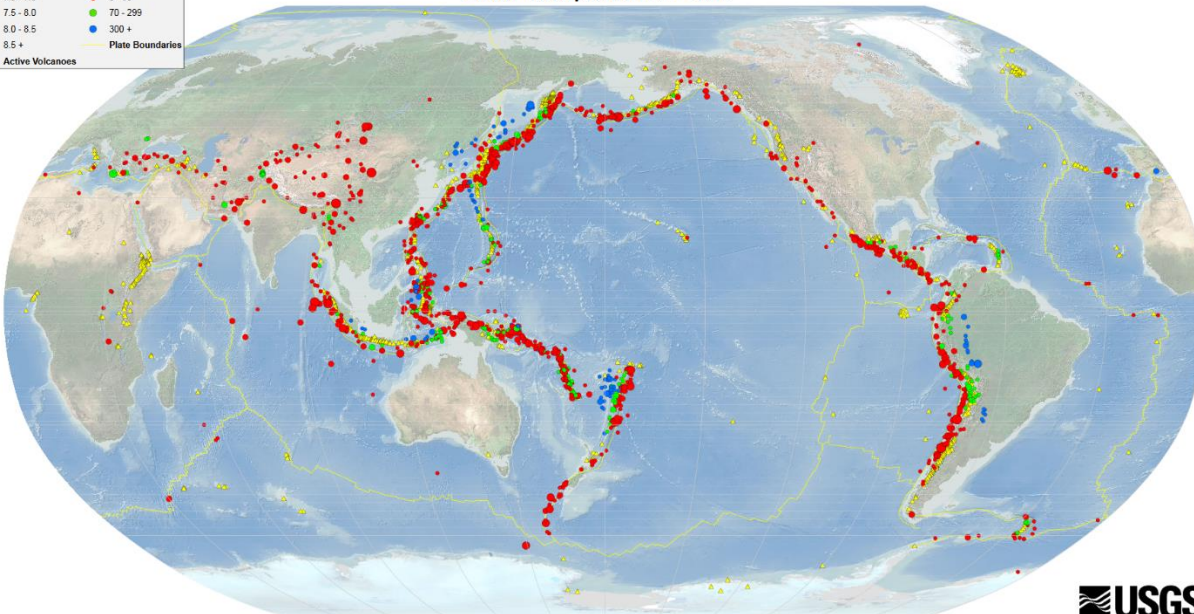


C.

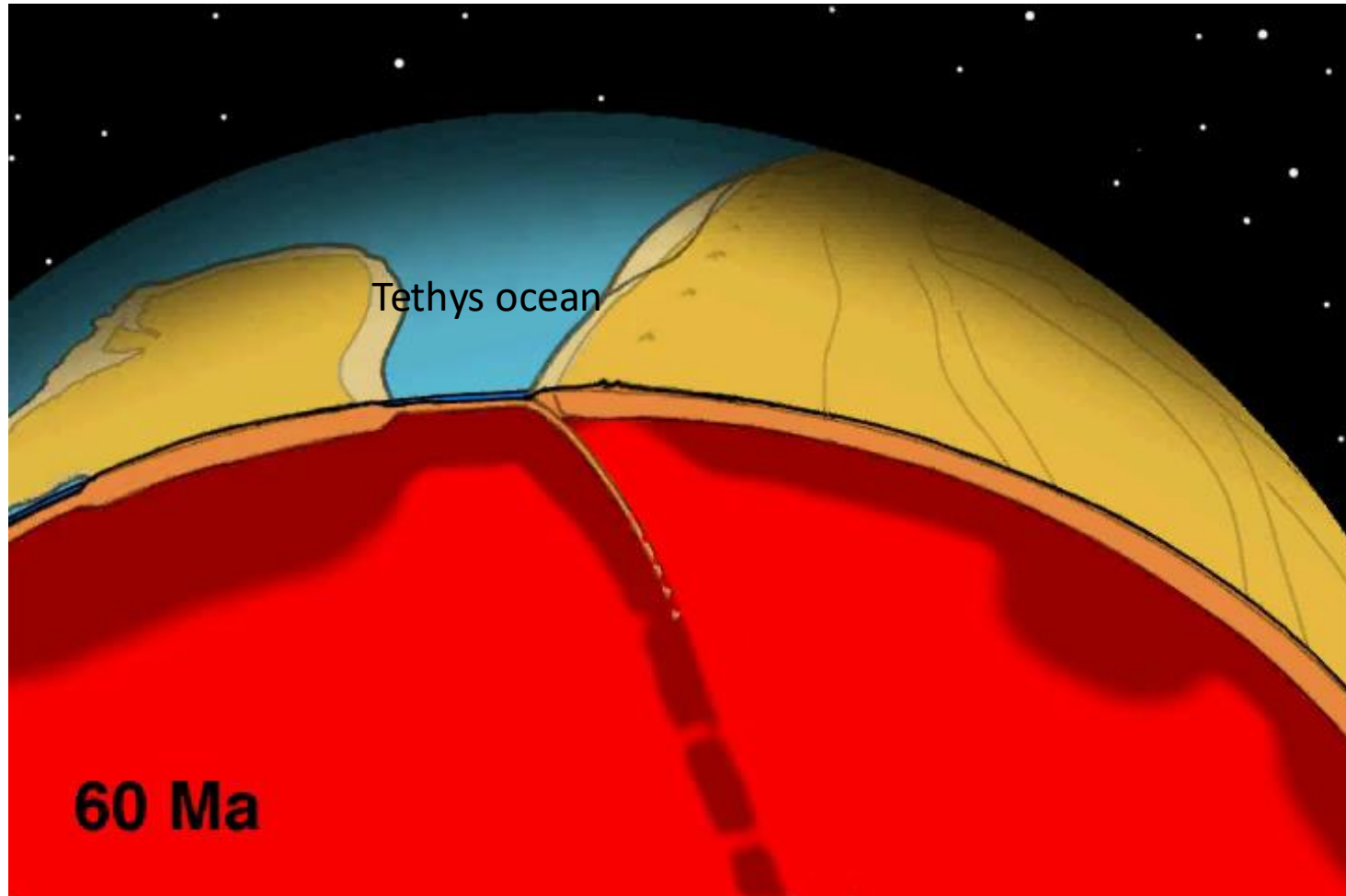
Earthquakes and volcanoes across the globe



Global Earthquakes 1900 - 2013



Formation of the mighty Himalayas (Tap to watch the video)



Marine fossils in the Himalayas!

